

U.S. Naval War College

## U.S. Naval War College Digital Commons

---

CMSI China Maritime Reports

China Maritime Studies Institute

---

2-24-2026

### China Maritime Report #51: The Seafarer Profession in Contemporary China

Conor M. Kennedy

Follow this and additional works at: <https://digital-commons.usnwc.edu/cmsi-maritime-reports>

---

#### Recommended Citation

Kennedy, Conor M., "China Maritime Report #51: The Seafarer Profession in Contemporary China" (2026). *CMSI China Maritime Reports*. 51.  
<https://digital-commons.usnwc.edu/cmsi-maritime-reports/51>

This China Maritime Report is brought to you for free and open access by the China Maritime Studies Institute at U.S. Naval War College Digital Commons. It has been accepted for inclusion in CMSI China Maritime Reports by an authorized administrator of U.S. Naval War College Digital Commons. For more information, please contact [repository.inquiries@usnwc.edu](mailto:repository.inquiries@usnwc.edu).



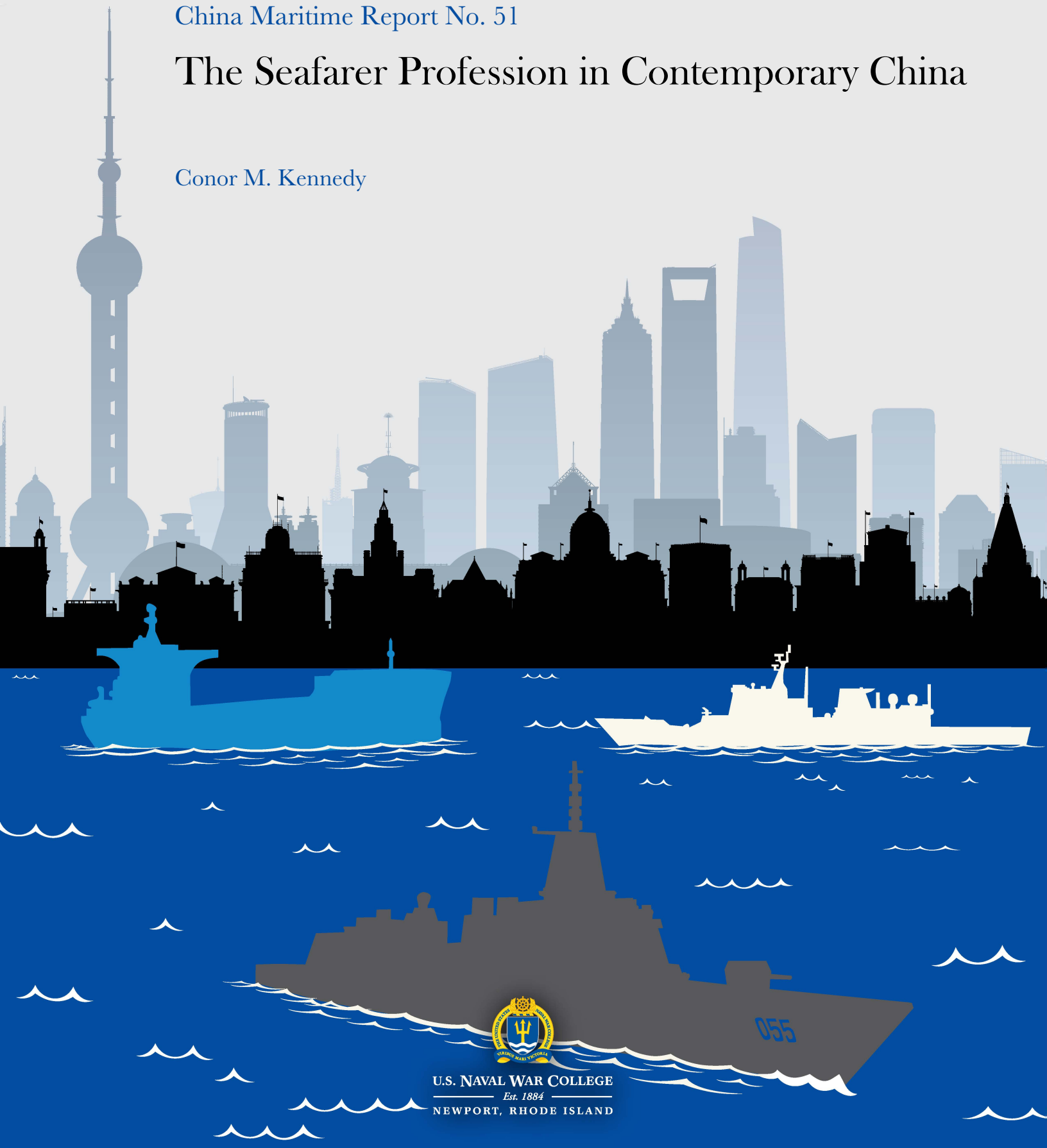
中国海事研究所  
China Maritime Studies Institute

February 2026

China Maritime Report No. 51

# The Seafarer Profession in Contemporary China

Conor M. Kennedy



U.S. NAVAL WAR COLLEGE  
Est. 1884  
NEWPORT, RHODE ISLAND

## Main Findings

- China's merchant marine is a foundational economic and national security asset, supporting the country's multi-decade growth and modernization and military logistics. Thus, Beijing views seafarers as a strategic resource and has prioritized development initiatives to strengthen the quality and quantity of the nation's professional merchant marine.
- China has a large potential labor supply of seafarers, with 943,954 registered for international and coastal shipping in 2023, but the actual number of seafarers actively working is far fewer. Over the past decade the sector has shifted from oversupply to emerging shortages due to COVID-19 disruptions, declining interest in maritime careers, workforce aging, and other factors. China will face continued challenges in sustaining its merchant marine workforce.
- Unique to China, some of its state-owned shipping fleets maintain a system of Ship Party Branches and ship political commissars to ensure the Chinese Communist Party is represented in the ship's leadership structure. Official regulations and laws do not explicitly acknowledge the presence of ship political commissars, but they nevertheless comprise an important part of the decision-making body aboard.
- Merchant marine careers are an attractive transition path for veterans who can qualify for officer licenses or rating positions through training and examinations. To assist with job placement and boost veteran entry into the workforce, the PRC government launched "Project Wave Break" in 2020, pairing up maritime schools, shipping companies, and authorities to facilitate this transition. These programs have expanded across China's provinces over the past several years.
- While veterans are traditionally sought after for service in the PRC's militia forces, the closer integration between training programs and companies by "Project Wave Break" has facilitated a pathway for veteran seafarers to join the maritime militia. Outside of major state-owned shipping enterprises, however, the lack of stable lifelong careers with companies is not conducive to the formation of militia organizations in enterprises.
- Despite improving regulation, information access, and standardization, China's mariner qualification and certification system faces persistent fraud and abuse, including falsified sea time, forged certificates, exam cheating, and data manipulation, which create hidden safety risks. There is significant fraud and abuse found in the many private crew service agencies that provide manning and crew management services for shipping companies. The Maritime Safety Administration conducts regular crackdowns, imposes fines, suspends licenses, and implements a national demerit points system, but has proven unable to fully eradicate the problem.
- COSCO Shipping's subsidiary COSCO Shipping Seafarer Management Co. Ltd. is the largest crew manning provider in China and has a robust in-house seafarer management system. In contrast, many companies outsource crew management to agencies that provide freelance seafarer labor. This has contributed to labor instability and other concerns, prompting the government to push firms to assume greater direct responsibility for their workforces.
- Despite major headwinds in the seafaring profession, the merchant marine remains a robust component of China's overall maritime power. Shipping and the merchant marine will continue to be central to the PRC's goal of becoming a "Transport Power" and the availability of commercial assets for military use will gradually improve.

## Introduction

The People's Republic of China (PRC) is one of the world's top seafarer suppliers, accounting for about one-third of the global workforce.<sup>1</sup> Its merchant marine has physically supported the PRC's rapid, multi-decade economic growth and continues to serve as a strategic workforce underpinning the nation's interaction with the global economy. This mammoth force enables the movement of key strategic inputs, such as crude oil, minerals, and other raw material imports, as well as countless other national outputs. Globally, seafarers are integral to supporting Beijing's geostrategic goals, such as regional shipping routes along the Maritime Silk Road.

The merchant marine is also critical for PRC national security and the People's Liberation Army's (PLA) strategic mobility. Under the national strategy of Military-Civil Fusion, China has created numerous ship transport units as part of the strategic delivery support forces (战略投送支援力量), including units in product tanker, bulk cargo, Roll-on/Roll-off ferries, salvage, and other various fleets of ships.<sup>2</sup> This effort seeks to provide the PLA with sufficient transport capacity in times of need, including both East Asian contingencies, such as an attack on Taiwan, and defense of China's "overseas interests."<sup>3</sup> China's merchant marine has historically been and will continue to be a critical manning resource in support of military seaborne logistics.<sup>4</sup>

Given the merchant marine's vital economic and military value, the PRC regards it as "an important strategic resource" that must be nurtured and developed.<sup>5</sup> Beijing has prioritized this critical workforce in many government documents and development initiatives.<sup>6</sup> This report examines the

---

<sup>1</sup> China was reported to be the second largest supplier of seafarers in the world in March 2025. See: Zahra Ahmed, "Top 10 Seafarer Supplying Nations in the World," *Marine Insight*, 25 March 2025, <https://www.marineinsight.com/know-more/top-10-seafarer-supplying-nations-in-the-world/>.

<sup>2</sup> Conor M. Kennedy, "Getting There: Chinese Military and Civilian Sealift in a Cross-Strait Invasion" in Ed. Joel Wuthnow, et al, *Crossing the Strait - China's Military Prepares for War with Taiwan*, National Defense University Press, Washington, D.C. (2022), pp. 231-233.

<sup>3</sup> Generally, China's shipping industry is legally required by the 2016 National Defense Transportation Law to render support to the military; however, preparations in training, modification, and organization conducted in peacetime make the activation of merchant shipping for national defense or national emergency far more efficient. 《中华人民共和国国防交通法》 ["Law of the People's Republic of China on National Defense Transportation"], 新华社 [*Xinhua*], 3 September 2016, [http://www.81.cn/gnxw/2016-09/03/content\\_7240929\\_5.htm](http://www.81.cn/gnxw/2016-09/03/content_7240929_5.htm).

<sup>4</sup> This is widely apparent in numerous PLA publications, including authoritative texts, regulations, and professional journals. Examples of these include: 张玉良 [Zhang Yuliang], ed., 战役学 [Science of Campaigns] (Beijing: National Defense Univ. Press, 2006), p. 303, 306; 肖天亮 [Xiao Tianliang], ed., 战略学 [Science of Military Strategy] (Beijing: National Defense Univ. Press, 2020), p. 260; 民用运力国防动员条例 ["Regulations on the Mobilization of Civilian Transportation Capacity for National Defense"], 中国政府法制信息网 [*China Government Legal Information Network*], 2 February 2004, [https://www.moj.gov.cn/pub/sfbgw/flfggz/flfggzxzf/200402/t20040202\\_350427.html](https://www.moj.gov.cn/pub/sfbgw/flfggz/flfggzxzf/200402/t20040202_350427.html); 李周清 [Li Zhouqing], 赵珍强 [Zhao Zhenqiang], 郭奎 [Guo Kui], and 王琦 [Wang Qi], 海上战略投送动员民船多点选型配置优化 ["Selection, Deployment and Optimization of Merchant Ships for Maritime Strategic Projection"], 军事交通学院学报 [Journal of Military Transportation University], no. 3 (2019), 4-7.

<sup>5</sup> 《中国船员发展规划（2016-2020年）》 ["Chinese Seafarer Development Plan (2016-2020)"], 交通运输部海事局发 (2016) 159 号 [Ministry of Transport Maritime Safety Administration Release (2016) No. 159], 2 September 2016, [https://www.gov.cn/gongbao/content/2017/content\\_5197028.htm](https://www.gov.cn/gongbao/content/2017/content_5197028.htm)

<sup>6</sup> Ibid. 国务院关于促进海运业健康发展的若干意见 ["Several Opinions of the State Council on Promoting the Healthy Development of the Maritime Industry"], 国务院 [State Council], Document No. 32, 2014, [https://www.gov.cn/zhengce/content/2014-09/03/content\\_9062.htm](https://www.gov.cn/zhengce/content/2014-09/03/content_9062.htm); 交通运输部教育部 财政部 人力资源和社会保障部 退役军人事务部 中华全国总工会关于加强高素质船员队伍建设的指导意见 ["The Ministry of Transport, Ministry of Education, Ministry of Finance, Ministry of Human Resources and Social Security, Ministry of Veterans Affairs, and the

fruits of these initiatives, providing an overview of the seafarer profession in contemporary China. It first explores workforce scale, structure, and sufficiency vis-à-vis industry requirements. It then details how seafarers are managed, trained and educated, and licensed to work aboard ships; how candidates join the merchant marine; and the roles of commercial enterprises in crew management. It probes persistent problems of fraud and abuse related to licensing and sea time, and in crew service agencies. Lastly, it attempts to determine whether China is achieving its seafarer development goals. In doing so, the report addresses trends impacting PRC seafarers as a workforce. The report finds that despite having a robust merchant marine and significant government support behind it, there are important long-term issues impacting the sustained development of this workforce.

## China's Shipping Industry

China leads the world in many indicators of maritime shipping power. It boasts one of the largest flagged merchant fleets at well over 12,000 vessels and 340 million dead weight tons (PRC and Hong Kong SAR combined) in January 2024. This includes the world's largest tanker fleet at 1,741 vessels (not including those registered to Hong Kong SAR), and large fleets of bulk and container vessels.<sup>7</sup> These figures do not include the large number of PRC-owned, foreign-flagged vessels operating internationally. This could be well over 2,000 vessels.<sup>8</sup> These figures also do not include the vast fleets of inland waterway vessels, numbering around 98,700.<sup>9</sup>

The bulk of China's shipping lies with state-owned enterprises (SOE) such as COSCO and China Merchants Group, accompanied by thousands of smaller private shipping enterprises. Of the 3,051 vessels listed in Figure 1 (below), 2,634 belong to state-owned shipping companies, excluding publicly-traded companies that are state-linked due to major SOE shares of ownership. While there is less information on state-owned shipping as a share of all Chinese shipping capacity, central and local SOEs dominate the list. Driven by an export-oriented economy, massive state-owned shipping conglomerates provide the strategic capacity to generate significant and sustained demand for manning by merchant mariners. This is supplemented by numerous private commercial enterprises,

---

All-China Federation of Trade Unions Issued Guiding Opinions on Strengthening the Development of High-Quality Seafarers”, 交通运输部 [Ministry of Transport], Document No. 41, 2021, chrome-extension://efaidnbmnmbpcjpcglclefindmkaj/https://xxgk.mot.gov.cn/xzgfxwj/202304/W020230428589089428107.pdf.

<sup>7</sup> “Merchant fleet by flag of registration and by type of ship, annual,” UN Trade & Development Data Hub, updated on 5 June 2024, <https://unctadstat.unctad.org/datacentre/dataviewer/US.MerchantFleet>; 王琛 [Wang Chen], 发挥海运在海洋强国建设中的重要作用 [“Give Full Play to the Important Role of Shipping in Building a Strong Maritime Nation”], 中国远洋海运 [China Ocean Shipping], No. 4, 2024,

[https://www.ndrc.gov.cn/xwdt/ztzl/NEW\\_srxxgcjjpijsx/yjcg/zw/202404/t20240419\\_1365787.html](https://www.ndrc.gov.cn/xwdt/ztzl/NEW_srxxgcjjpijsx/yjcg/zw/202404/t20240419_1365787.html); 中国第一! 全球十大船东国最新排名出炉 [“China is No. 1! The Latest Ranking of the World's Top Ten Shipowners is Released”], 国际船舶网 [eworldship.com], 26 January 2025,

[https://www.eworldship.com/html/2025/ship\\_market\\_observation\\_0126/209728.html](https://www.eworldship.com/html/2025/ship_market_observation_0126/209728.html). Authoritative sources provide some sense of the scale of state-owned shipping; however, this gets significantly more complicated in companies not directly owned by the state, but are instead state-controlled or influenced through ownership structures.

<sup>8</sup> In 2012, there were 1609 Chinese-owned ships flying flags of convenience, which possibly grew to a reported 2791 by the end of 2023, despite long-term desire by China to entice these companies to fly the PRC flag. 40 名上海政协委员呼吁: 出台政策吸引中资“方便旗船”回归 [“40 Shanghai CPPCC Members Call for Policies to Attract Chinese-funded ‘Flag of Convenience Ships’ to Return”], 财新 [Caixin], 24 January 2024, <https://companies.caixin.com/2024-01-24/102160117.html>; 上海率先试点保税船舶登记 吸引中资方便旗船回归 [“Shanghai Takes the Lead in Piloting Bonded Ship Registration to Attract Chinese-funded Flag of Convenience Ships to Return”], 新华网 [Xinhua], 6 March 2012, <https://sisi.shmtu.edu.cn/2012/0306/c8822a91361/page.htm>.

<sup>9</sup> 2024 年交通运输行业发展统计公报 [“Statistical Bulletin on the Development of the Transportation Industry in 2024”], 交通运输部 [Ministry of Transport], 12 June 2024, [https://www.gov.cn/lianbo/bumen/202506/content\\_7027415.htm](https://www.gov.cn/lianbo/bumen/202506/content_7027415.htm).

of which many publicly-traded companies are state-linked or state-controlled in their ownership structures.<sup>10</sup>

Figure 1: Major PRC Shipping Companies by Vessel Capacity as of 31 December 2024<sup>11</sup>

| # | Company Name (Chinese/English)                                          | Type                              | Estimated Fleet Size / DWT   |
|---|-------------------------------------------------------------------------|-----------------------------------|------------------------------|
| 1 | COSCO Shipping Group<br>(中国远洋海运集团有限公司)                                  | Central SOE                       | 1,535 ships / 128.96 million |
| 2 | China Merchants Group<br>(招商局集团有限公司)                                    | Central SOE                       | 546 ships / 44.437 million   |
| 3 | Shandong Shipping Corporation<br>(山东海运股份有限公司)                           | Local SOE, Shandong <sup>12</sup> | 81 ships / 10.863 million    |
| 4 | Bohai Shipping (Hebei) Co., Ltd.<br>(渤海远洋(河北)运输有限公司)                    | Local SOE, Hebei <sup>13</sup>    | 42 ships / 4.729 million     |
| 5 | Fujian Guohang Ocean Shipping (Group) Co., Ltd.<br>(福建国航远洋运输(集团)股份有限公司) | Private <sup>14</sup>             | 62 / 4.202 million           |
| 6 | Zhejiang Xiehai Group Co., Ltd.<br>(浙江协海集团有限公司)                         | Private <sup>15</sup>             | 52 / 4.071 million           |
| 7 | Fujian Highton Development Co., Ltd.<br>(福建海通发展股份有限公司)                  | Private <sup>16</sup>             | 65 / 3.776 million           |

<sup>10</sup> Discussion on the complexity of state-influence over private enterprises can be found in many studies. For example, several authors found that the number of state-connected private enterprises has expanded over the last 20 years. See Chong-En Bai, Chang-Tai Hsieh, Zheng Michael Song, and Xin Wang, “The Rise of State-Connected Private Owners in China,” National Bureau of Economic Research – NBER Working Paper Series (Working Paper 28170), December 2020, pp. 2-3, p. 6-8.

<sup>11</sup> 关于 2024 年航运企业船舶运力排名信息的公示 [“Announcement Regarding the Ranking of Shipping Companies by Vessel Capacity in 2024”], 上海航运交易所 [Shanghai Shipping Exchange], 11 April 2025, [https://www.sse.net.cn/cninfo/notice/policy/202504/t20250411\\_1402698.jsp](https://www.sse.net.cn/cninfo/notice/policy/202504/t20250411_1402698.jsp).

<sup>12</sup> The Shandong provincial government controls Shandong Marine Group (山东海洋集团), which has a controlling stake in the Shandong Shipping Corporation. 公司简介 [“Company Introduction”], 山东海运股份有限公司 [Shandong Shipping Corporation], undated, <https://www.sdshipping.cn/column/1.html>.

<sup>13</sup> 向海发展启新程 向海图强逐深蓝—渤海远洋打造中国海运力量新一极 [“Embarking on a New Journey Towards Maritime Development, Striving for Strength on the Seas – Bohai Ocean Shipping is Building a New Level of Power in China’s Maritime Industry”], 河北经济日报 [Hebei Economic Daily], 22 August 2025, <https://www.xindemarinenews.com/china/61042.html>.

<sup>14</sup> Chen Yang, “Guohang Ocean Shipping Sets Up Subsidiary in Singapore to Accelerate Global Expansion,” Xinde Marine News, 27 June 2025, <https://www.xindemarinenews.com/en/carrier/2025/0627/60248.html>.

<sup>15</sup> 浙江协海集团有限公司 [“Zhejiang Xiehai Group Co. Ltd.”], 台州现代物流网 [www.tzxdwl.com], 20 March 2020, <http://www.tzxdwl.com/qiyexinxi/2020-03-20/2296.html>.

<sup>16</sup> 公司概况 [“Company Profile”], 海通发展 [Highton Development], undated, <http://www.fj-highton.com/default.aspx?pageid=99>.

| #  | Company Name (Chinese/English)                             | Type                                       | Estimated Fleet Size / DWT |
|----|------------------------------------------------------------|--------------------------------------------|----------------------------|
| 8  | Zhonggu Shipping Group Co., Ltd.<br>(中谷海运集团有限公司)           | Private <sup>17</sup>                      | 111 / 3.717 million        |
| 9  | Zhejiang Shipping Group Co., Ltd.<br>(浙江省海运集团股份有限公司)       | Local SOE, Zhejiang <sup>18</sup>          | 52 / 3.26 million          |
| 10 | Antong Holdings Co., Ltd.<br>(安通控股股份有限公司)                  | Mixed <sup>19</sup>                        | 85 / 2.543 million         |
| 11 | Shanghai Leading Energy Shipping Co., Ltd.<br>(上海瑞宁航运有限公司) | Central SOE <sup>20</sup>                  | 32 / 2.438 million         |
| 12 | Guoneng Ocean Shipping Co., Ltd.<br>(国能远海航运有限公司)           | Central SOE <sup>21</sup>                  | 40 / 2.17 million          |
| 13 | Jiangsu Ocean Shipping Co., Ltd.<br>(江苏远洋运输有限公司)           | Local SOE, Jiangsu <sup>22</sup>           | 120 / 1.807 million        |
| 14 | Shanghai Time Shipping Co., Ltd.<br>(上海时代航运有限公司)           | Central SOE <sup>23</sup>                  | 28 / 1.663 million         |
| 15 | Ningbo Ocean Shipping Co., Ltd.<br>(宁波远洋运输股份有限公司)          | Local SOE, Zhejiang <sup>24</sup>          | 100 / 1.558 million        |
| 16 | Ningbo Marine Co., Ltd.<br>(宁波海运股份有限公司)                    | Local SOE, Zhejiang Province <sup>25</sup> | 31 / 1.529 million         |

<sup>17</sup> “Enterprise Profile,” *Zhonggushipping.com*, undated, <http://www.zhonggushipping.com/About/About>.

<sup>18</sup> 浙江省海运集团招聘公告 [“Zhejiang Shipping Group Recruiting Announcement”], 信德海事网 [*Xinde Marine News*], 11 October 2022, <https://www.xindemarinenews.com/job/zhaopin/2022/1011/42434.html>.

<sup>19</sup> Various China Merchant Groups companies own majority stakes in Antong Holdings. 安通控股十大股东一览 [“List of Antong Holding’s Top Ten Shareholders”], 东方财富网 [*eastmoney.com*], 30 September 2025, <https://data.eastmoney.com/gdxf/stock/600179.html>.

<sup>20</sup> 上海瑞宁航运有限公司 [“Shanghai Leading Energy Shipping Co. Ltd.”], 企查查 [*Qcc.com*], undated, <https://www.qcc.com/firm/ad385c6239b12efcb0f9af38b781b0ed.html>, accessed 2 January 2026.

<sup>21</sup> 国家能源集团航运发展公司正式成立 [“China Energy Group Shipping Development Company Has Been Officially Established”], 中华航运 [*ChineseShipping.com.cn*], 20 December 2022, [https://info.chineseshipping.com.cn/cninfo/News/202212/t20221220\\_1372585.shtml?utm\\_source=chatgpt.com](https://info.chineseshipping.com.cn/cninfo/News/202212/t20221220_1372585.shtml?utm_source=chatgpt.com).

<sup>22</sup> 江苏远洋: 点燃“强引擎” 驶入“新航程” [“Jiangsu Ocean Shipping: Igniting a ‘Powerful Engine’ and Embarking on a ‘New Voyage’”], 江苏省政府国有资产监督管理委员会 [*Jiangsu Provincial Government State-Owned Assets Supervision and Administration Committee*], 25 May 2021, [https://jsgzw.jiangsu.gov.cn/art/2021/5/25/art\\_11717\\_9826253.html](https://jsgzw.jiangsu.gov.cn/art/2021/5/25/art_11717_9826253.html).

<sup>23</sup> 上海时代航运有限公司 [“Shanghai Times Shipping Co., Ltd.”], 中远海运散货运输有限公司 [*COSCO Shipping Bulk Co. Ltd.*], undated, [https://bulker.coscoshipping.com/col21736/art/2021/art\\_21736\\_214264.html?utm\\_source=chatgpt.com](https://bulker.coscoshipping.com/col21736/art/2021/art_21736_214264.html?utm_source=chatgpt.com), accessed 6 January 2026.

<sup>24</sup> Owned by the provincial SOE Ningbo Zhoushan Port Company Ltd. (宁波舟山港股份有限公司). 又一国企航运公司进军外贸散货市场! [“Another State-Owned Shipping Company is Entering the Foreign Trade Bulk Cargo Market!”], 海运圈聚焦 [*CNSS.com*], 7 August 2024, <https://www.cnss.com.cn/html/hyqy/20240807/354280.html>.

<sup>25</sup> 宁波海运股份有限公司 [“Ningbo Marine Co., Ltd.”], 大连海事大学 – 就业信息网 [*Dalian Maritime University – Career Information Network*], 12 September 2024, <https://myjob.dlmu.edu.cn/teachin/view/id/149107>.

| #                                                                                                                                                                                                                                                                                                                                                                                                     | Company Name (Chinese/English)                        | Type                      | Estimated Fleet Size / DWT     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------|--------------------------------|
| 17                                                                                                                                                                                                                                                                                                                                                                                                    | Dongguan Haichang Shipping Co., Ltd.<br>(东莞市海昌船务有限公司) | Private <sup>26</sup>     | 18 / 1.477 million             |
| 18                                                                                                                                                                                                                                                                                                                                                                                                    | Huayuan Star Shipping Co., Ltd.<br>(华远星海运有限公司)        | Central SOE <sup>27</sup> | 27 / 1.317 million             |
| 19                                                                                                                                                                                                                                                                                                                                                                                                    | Shanghai Beihai Shipping Co., Ltd.<br>(上海北海船务股份有限公司)  | Mixed <sup>28</sup>       | 24 / 1.315 million             |
|                                                                                                                                                                                                                                                                                                                                                                                                       | <b>TOTAL</b>                                          |                           | <b>3,051 / 225.832 million</b> |
| <b>Notes:</b><br>Central SOE – State-owned enterprise controlled by the central government via the State-owned Assets Supervision and Administration Commission of the State Council (SASAC).<br>Local SOE – State-owned enterprise controlled by provincial or municipal governments.<br>DWT – Deadweight tonnage (measure of a ship’s carrying capacity excluding its own structure and machinery). |                                                       |                           |                                |

## Ship Organization

The organizational system aboard PRC merchant ships is similar to international practices, with the addition of some Chinese Communist Party (CCP) elements. As with other nations’ vessels, there is a deck department (舱面部 or 甲板部), engine department (轮机部 or 机舱部), and the steward department (业务部).<sup>29</sup> The deck department oversees navigation, cargo handling, and management; the engine department is responsible for vessel propulsion, maintenance, and equipment; and the steward department manages food, sanitation, and accommodations for the crew. Figure 2 (below) situates each ship position and its equivalent term in English. The ship hierarchy is essential to ensuring command and operation of any vessel, with ultimate authority belonging to the captain.<sup>30</sup>

<sup>26</sup> “Company Profile,” 东莞海昌 [Dongguan Haichang], undated, <https://www.dghaichang.com/en/list/148.html>, accessed 10 January 2026.

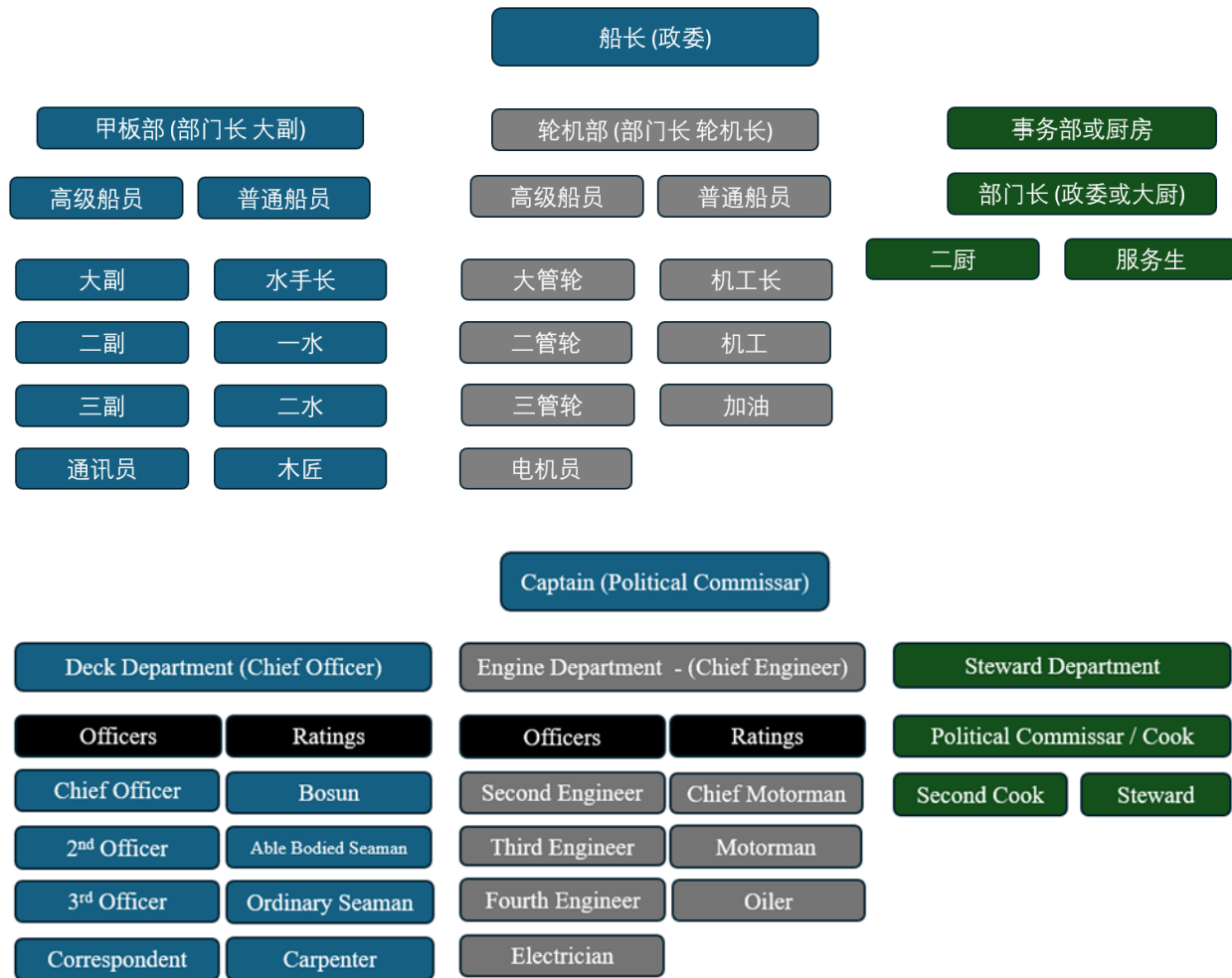
<sup>27</sup> Huayuan Star Shipping is under the Central SOE China Huadian Corporation Ltd. (中国华电集团有限公司). 启航! 华远星海运在天津港保税区成立全资子公司发展航运业务 [“Setting Sail! Huayuan Xinghai Shipping Establishes a Wholly-Owned Subsidiary in the Tianjin Port Free Trade Zone to Develop its Shipping Business”], 中国(天津)自由贸易试验区天津机场片区 - 保税区党建工作部 [Tianjin Airport Area of China (Tianjin) Pilot Free Trade Zone – Bonded Zone Party Building Work Department], 26 September 2023, <https://www.tjftz.gov.cn/contents/6097/363208.html>.

<sup>28</sup> Shanghai Beihai Shipping is not itself an SOE, but its largest shareholders include major SOEs such as COSCO Shipping Energy Transportation Co. Ltd. (中远海运能源运输股份有限公司) and China National Offshore Oil Corporation’s CNOOC Petrochemical Import & Export Co. Ltd. 上海北海船务股份有限公司需求信息 [“Recruitment Information from Shanghai Beihai Shipping Co., Ltd.”], 大连海事大学就业信息网 [Dalian Maritime University Career Information Network], 20 October 2021, <https://myjob.dlmu.edu.cn/news/view/aid/352229/tag/qzjq>.

<sup>29</sup> 徐丘濂 [Xu Qiulian], 齐玉霞 [Qi Yuxia], and 林景余 [Lin Jingyu], 扬帆远航 他们在“流动的国土”上守“土”尽责 [“Setting Sail for a Long Journey, They are Doing Their Duty Aboard the ‘Moving Territory’”], 南方工报 [Southern Worker’s Daily], 11 July 2023, [http://www.nfgb.com.cn/nggc/content/post\\_1081610.html](http://www.nfgb.com.cn/nggc/content/post_1081610.html).

<sup>30</sup> Article 21, 《中华人民共和国船员条例》(2023 修订) [“Regulation of the People’s Republic of China on Seamen (2023 Revision)”], 国务院令 第 494 号 [State Council Order No. 494], 20 July 2023, <https://www.msa.gov.cn/page/article.do?type=hsfg&articleId=C60C35B6D6C64C0EB67730EFB589D720>.

Figure 2: Typical Ship Positions in Chinese & English<sup>31</sup>



By law, Chinese-flagged ships must be captained by PRC citizens.<sup>32</sup> The captain is in command of the vessel and its departments, overseeing the overall operations of the vessel and its crew. According to the “Regulation of the People’s Republic of China on Seamen” (《中华人民共和国船员条例》), the captain has independent decision-making power and bears the ultimate responsibility for ensuring the safety of life and property on the water, maintaining ship security, and preventing ship pollution. The captain is supported by management and operations-grade officers that carry out ship operations and oversee the ratings.<sup>33</sup>

<sup>31</sup> 所有职位船员的在船职责 [“Onboard Duties of All Crew Members”], 信德海事 [Xinde Marine News], 21 July 2017, <https://www.xindemarineneeds.com/data/hanghaijishu/286.html>.

<sup>32</sup> Article 8 of “Regulation of the People’s Republic of China on Seamen (2023 Revision).”

<sup>33</sup> For a complete official description of the captain’s role and responsibilities, see Articles 17-20 of 《中华人民共和国船员条例》(2023 修订) [“Regulation of the People’s Republic of China on Seamen (2023 Revision)”], 国务院令 第 494 号 [State Council Order No. 494], 20 July 2023, <https://www.msa.gov.cn/page/article.do?type=hsfg&articleId=C60C35B6D6C64C0EB67730EFB589D720>.

Captains, officers and ratings are further classified by the requirements of their positions, divided into management-grade (“M” or 管理级), operations-grade (“O” or 操作级), and support-grade (“S” or 支持级) roles.<sup>34</sup> Each ship position is classified according to the following.

**Management-grade:** Captain, political commissar, chief officer, chief engineer, second engineer.

**Operations-grade:** Second officer, third officer, third engineer, fourth engineer, electrician.

**Support-grade:** Bosun, able-bodied seaman, ordinary seaman, chief motorman, motorman, carpenter, cook.<sup>35</sup>

The PRC’s merchant marine officers (高级船员) and ratings (普通船员) are largely aligned with international norms in terms of specific ranks and duties. While officer ranks are fairly straightforward, the ratings, or support-grade personnel, merit some explanation because they are not intuitive to those unfamiliar with merchant marine careers. Among the ratings, there are entry-level ordinary seaman (OS, 二级水手), who are starting their careers and perform various deck work and cargo handling functions under the direction of the senior rating onboard. After gaining experience and training, an OS may upgrade to become an able-bodied seaman (AB, 一级水手). From there, the rank of a rating depends on their experience and qualification to participate in a navigation or engine watch. Watch standing is a critical aspect of ship operation and must be maintained at all times. While officers are qualified to organize and keep a watch as the Officer in Charge of the Navigational Watch (OICNW, 驾驶员), they must work in shifts and are assisted by ratings that qualify to participate in watch standing duties.<sup>36</sup> With sufficient experience, training, and passing an exam, an AB can become a “rating forming part of a navigational watch” (RFPNW, 值班水手) or a “rating forming part of an engine watch” (RFPEW, 值班机工). This qualification allows a rating to join a navigational watch and perform watch standing duties such as maintaining a helm watch or standing lookout. From there, a rating may advance to become an able seafarer – deck (AS-D, 高级值班水手) or able seafarer – engine (AS-E, 高级值班机工). Lastly, the most senior rating aboard a ship is referred to as the boatswain or bosun (水手长), who oversees the work of the ratings in the deck department. There is an equivalent senior rating in the engine department, called the chief motorman (机工长).<sup>37</sup>

Officers in the deck and engine departments are more senior management or operations-grade seafarers with greater responsibilities. Officers begin as either third officer or fourth engineer and

---

<sup>34</sup> Article 9 of 《中华人民共和国海船船员适任考试和发证规则》 [“Regulations on Examination and Certification of Competence for Seafarers of the People’s Republic of China”], 交通运输部令 2020 年第 11 号 [Order of the Ministry of Transport 2020, No. 11], 7 July 2020, [https://www.gov.cn/zhengce/zhengceku/2020-07/14/content\\_5526802.htm](https://www.gov.cn/zhengce/zhengceku/2020-07/14/content_5526802.htm).

<sup>35</sup> 船员职务有哪些 [“What Are the Crew Duties?”], 中国水运报 [China Water Transport], 24 January 2024, <https://www.zgsyb.com/news.html?aid=673107>.

<sup>36</sup> The chief officer, second officer, third officer are all considered OICNW or 驾驶员 according to Article 7 of “Regulations on Examination and Certification of Competence for Seafarers of the People’s Republic of China.”

<sup>37</sup> These positions and their requirements are outlined in the attached requirements entitled 申请海船船员适任证书的培训、海上任职资历和适任考试要求 [“Training, Sea Service Qualifications and Proficiency Examination Requirements for Applying for a Seafarer’s Certificate of Competency”] in “Regulations on Examination and Certification of Competence for Seafarers of the People’s Republic of China”; 船员入行科普：不同水手职位都有什么区别? [“Popular Knowledge for Seafarers: What are the Difference Between Different Sailor Positions?”], 航运 e 家 [hyej.net], 9 March 2022, [https://www.hyej.net/cms/detail\\_80.html](https://www.hyej.net/cms/detail_80.html).

advance upwards toward more senior officer positions. Ultimately, they can become captain or chief engineer of a vessel when qualified.

Merchant mariner qualifications will be discussed in greater detail in later sections, but it is important to note here the three navigational zones in which they may be licensed to work. Navigational zones are divided into “unlimited navigation” (无限航区), “coastal navigation” (沿海航区), and “inland waterways” (内河). Unlimited navigation refers to any navigable waters at sea, including open ports and international navigable canals and rivers in any country. Coastal navigation refers to ports, inland waterways, territorial waters, and all other navigable waters under PRC jurisdiction.<sup>38</sup> The different zones act as restrictions on a mariner’s license. A mariner licensed only for inland waterways would be restricted to just that zone and would need to qualify for zone expansion.

Lastly, merchant mariners are sometimes referred to by the type of labor contract under which they work. Stated simply, traditional labor contracts between seafarers and shipping companies or shipowners are referred to as “in-house seafarers” (自有船员) while others rely on a labor contract established between seafarers and a crew service agency that provides manning services for shipping companies, often referred to as “freelance seafarers” (自由船员). The latter category is more fluid in the labor market as they typically do not see themselves as employees of the major shipping companies, simply hired labor.<sup>39</sup>

Unique to Chinese merchant shipping, the CCP also maintains a system of control over China’s state-owned oceangoing merchant fleets through Party branches and ship political commissars. Mention of this system is absent from most relevant regulations and laws pertaining to China’s merchant marine. Nonetheless, the ship Party branch serves as an important decision-making body and leadership structure on board. It is composed of the “Three Chiefs” (三长), i.e., the three senior officers aboard the vessel. The Party Branch is led by the Party secretary, a position held by the ship’s political commissar. The ship captain serves as the deputy secretary and the chief engineer as a branch member.<sup>40</sup> These individuals likely provide their own perspectives in ship decision-making, as it pertains to their professional roles, but it is unknown how big a role the Party branch plays in specific planning of ship operations.

The political commissar acts as a representative of the CCP and serves as secretary of the Party Branch established on ships operated by state-owned shipping conglomerates. Political commissars historically held prominent positions of authority in PRC ocean shipping and were briefly in

---

<sup>38</sup> Article 63, 《中华人民共和国海船船员适任考试和发证规则》 [“Rules for the Examination and Licensing of Seafarers of the People’s Republic of China”], 交通运输部网站 [Ministry of Transport Website], 26 April 2022, [https://www.gov.cn/zhengce/2022-04/26/content\\_5717878.htm](https://www.gov.cn/zhengce/2022-04/26/content_5717878.htm).

<sup>39</sup> There are more complicated labor arrangements, but they broadly fall under these two categories. 蔡滢炜 [Cai Yingwei] and 何东闽 [He Dongmin], 船员劳动法律关系辨析 [“Analysis of the Labor Legal Relationship of Seafarers”], 金诚同达 [Jincheng, Tongda & Neal], 11 April 2018, [https://www.jtn.com/CN/booksdetail.aspx?type=06001&keyid=0000000000000003448&PageUrl=majorbook&Lan=CN](https://www.jtn.com/CN/booksdetail.aspx?type=06001&keyid=0000000000000003448&PageUrl=majorbook&Lan=CN;); According to one article posted in 2024, around 70 percent of seafarers are considered some form of freelance labor, working without labor contracts. 又内卷又缺人？船员市场正陷入奇怪趋势！ [“Increasingly Competitive While Also Having Shortages? The Crew Market is Experiencing a Strange Trend!”], 航运e家 [hyej.net], 25 September 2024, [https://www.hyej.net/cms/detail\\_519.html](https://www.hyej.net/cms/detail_519.html).

<sup>40</sup> 中国海运表彰长期服务航运事业的船舶“三长” [“China Shipping Honors ‘Three Chiefs’ That Have Served The Shipping Industry for a Long Time”], 中国海运(集团)总公司 [China Shipping (Group) Company], 6 August 2014, <http://www.sasac.gov.cn/n2588025/n2588124/c3783093/content.html>.

command of vessels with authorities greater than the captain from 1975 to 1978 as a result of the Cultural Revolution. The position originated out of military practice to ensure Party-state control over the seagoing merchant fleet. Since then, political commissars' significance and prestige has declined under the tide of modernization and industry and market pressures. The practice of assigning political commissars to ships is still implemented today and, in some cases, has received recent strengthening. When visiting foreign ports, they are not listed on crew manifests, at least not as political commissars. Typically, they will be listed as one of the ratings, usually an ordinary seaman.<sup>41</sup> Their role is primarily to assist the captain, carry out political and administrative work in the management of ship crews, and provide many other supporting functions. The political commissar acts as the second-most senior officer on the ship and, like the captain, is generally above the ship departments.<sup>42</sup> The ship political commissar can assert some influence over personnel decisions via a work performance and post-voyage reporting process with companies.<sup>43</sup>

## Labor Supply and Demand

The PRC is one of the world's leading suppliers of seafarers, alongside countries like the Philippines, Ukraine, India, and Russia, fulfilling both international and domestic maritime labor needs. By the end of 2023, the PRC had just under two million "registered mariners" (注册船员). This large figure is an aggregate counting of all merchant marine trades across various industries in China, including crew registered to work in inland waterways, coastal routes, and international routes. It does not include the 143,582 "dispatched seafarers" (外派海员) reported that year.<sup>44</sup> This term refers to Chinese seafarers hired to work aboard foreign-owned vessels, also including vessels registered to Hong Kong, Macau, and Taiwan. The total number of registered mariners also does not include the nearly 16 million people working in the fishing industry nationally in 2023.<sup>45</sup> The total figures of registered mariners has continued to gradually expand over the past decade. These large figures are frequently noted publicly, but are not reflective of the actual active workforce, as will be explored below.

---

<sup>41</sup> 赵明华 [Zhao Minghua], 石秀印 [Shi Xiuyin], and 冯同庆 [Feng Tongqing], 海洋上的社会 – 中国商船政委与船员的航海生活 [*The Ship Political Commissar and His Shipmates – Transformation of the Commissar's Role on Chinese Ships*], (Beijing: Social Services Documentation Publishing House, 2004), p. 15.

<sup>42</sup> For greater detail on PRC ship political commissars, see: Conor M. Kennedy, "Onboard Political Control: The Ship Political Commissar in Chinese Merchant Shipping," China Maritime Studies Institute, August 2024, <https://digital-commons.usnwc.edu/cmsi-maritime-reports/40/>; 所有职位船员的在船职责 ["Onboard Duties of All Crew Positions"], 信德海事 [*Xinde Marine News*], 21 July 2017, <https://xindemarinenews.com/data/hanghaijishu/286.html>.

<sup>43</sup> Specifically, ship personnel, including the captain, will be reviewed on their individual work performance, which is submitted by the political commissar in their personnel administrative capacity. See: Zhao, Shi, and Feng, *The Ship Political Commissar and His Shipmates – Transformation of the Commissar's Role on Chinese Ships*, pp. 48-49, 54-56, 92.

<sup>44</sup> 2023 年中国船员发展报告 ["2023 Annual Report on Chinese Seafarer Development"], June 2024, pp. 1-3.

<sup>45</sup> 2023 年全国渔业经济统计公报 ["2023 National Fisheries Economic Statistical Bulletin"], 农业农村部渔业渔政管理局 [*Ministry of Agriculture and Rural Affairs Fisheries Administration*], 5 July 2024, [http://www.yyj.moa.gov.cn/gzdt/202407/t20240705\\_6458486.htm#:~:text=%E5%85%AD%E3%80%81%E6%B8%94%E4%B8%9A%E4%BA%BA%E5%8F%A3%E5%92%8C%E6%B8%94%E4%B8%9A,%E4%B8%87%E4%BA%BA%E3%80%81%E4%B8%8B%E9%99%8D1.73%25%E3%80%82](http://www.yyj.moa.gov.cn/gzdt/202407/t20240705_6458486.htm#:~:text=%E5%85%AD%E3%80%81%E6%B8%94%E4%B8%9A%E4%BA%BA%E5%8F%A3%E5%92%8C%E6%B8%94%E4%B8%9A,%E4%B8%87%E4%BA%BA%E3%80%81%E4%B8%8B%E9%99%8D1.73%25%E3%80%82). Perhaps more relevant to this research, there were over 5.25 million people engaged in marine fishing in 2020, including traditional fisherfolk and those involved in industrial fishing operations. 中国渔业总体产业规模庞大, 渔民人口数量高达 1721 万人, 广东渔民人口 216 万人, 居全国之首 ["China's Fishery Industry Is Large Scale with a Fishing Population of 17.21 million. Guangdong Has a Fishing Population of 2.16 Million, Ranking First in the Country"], 智研咨询 [*Thinktank Consulting*], 24 December 2021, <https://www.chyxx.com/industry/202112/990548.html>.

Figure 3: Selected Years of PRC Registered Merchant Marine<sup>46</sup>

|             | Int'l Route      | Coastal Route    | Inland Waterways  | Dispatched       |
|-------------|------------------|------------------|-------------------|------------------|
| <b>2014</b> | 447,054 (+6.69%) | 161,413 (+4.08%) | 708,000           | 125,000          |
| <b>2015</b> | 470,512 (+5.25%) | 168,478 (+4.38)  | 731,234 (+3.29)   | 133,326 (+7.28)  |
| <b>2016</b> | 497,297 (+5.69%) | 175,764 (+4.32%) | 719,790           | 142,738 (+7.06)  |
| <b>2017</b> | 524,498 (+5.46%) | 184,524 (+4.98%) | 774,225           | 138,854 (-2.7%)  |
| <b>2018</b> | 545,877 (+4.07%) | 191,780 (+3.93%) | 837,577           | 145,922 (+5.1%)  |
| <b>2019</b> | 575,823 (+5.48%) | 208,532 (+8.75%) | 874,833 (+4.4%)   | 155,449 (+6.5%)  |
| <b>2020</b> | 592,998 (+7.94%) | 215,185 (+3.19%) | 908,683           | 122,304 (-21.3%) |
| <b>2021</b> | 620,519 (+4.64%) | 230,041 (+6.90%) | Unavailable       | Unavailable      |
| <b>2022</b> | 654,775 (+5.52%) | 245,949 (+6.92%) | 996,353           | 127,000          |
| <b>2023</b> | 682,635 (+4.25%) | 261,319 (+6.25%) | 1,038,209 (+4.2%) | 143,582 (+13.3)  |

The numbers in Figure 3 (above) encompass all registered mariners and are only distinguished by their navigational zones, which correspond with the navigational zone of mariner licenses. It does not break them down into their ranks, qualifications, vessel class, and many other categories.

Importantly, it does not actually count those that worked aboard vessels at sea and logged “approved sea service” (海上服务资历) and thus represents a *potential* labor supply, not those actively working at sea in a given year.<sup>47</sup> Moreover, there are only four circumstances in which a registration will be cancelled, including death or disappearance, loss of civil capacity, legal revocation of seafarer service records, or an individual application for cancellation of registration. Since these circumstances are

<sup>46</sup> These figures are drawn from the Information Office of the PRC Ministry of Transport (中华人民共和国交通运输部新闻办公室): 2016 年中国船员发展报告 [“2016 Annual Report on Chinese Seafarer Development”], June 2017, [http://www.360doc.com/content/17/0624/17/35131417\\_666215529.shtml](http://www.360doc.com/content/17/0624/17/35131417_666215529.shtml); 2017 年中国船员发展报告 [“2017 Annual Report on Chinese Seafarer Development”], June 2018, pp. 3-6; 2018 年中国船员发展报告 [“2018 Annual Report on Chinese Seafarer Development”], June 2019, pp. 3-6; 2020 年中国船员发展报告 [“2020 Annual Report on Chinese Seafarer Development”], June 2021, pp. 3-11; 2023 年中国船员发展报告 [“2023 Annual Report on Chinese Seafarer Development”], June 2024, pp. 3-6.

<sup>47</sup> Approved sea service typically looks back five years to prove a mariner’s training, examination, certification, and safety record. If a mariner is primarily responsible for a major accident and is subject to penalties by the Maritime Safety Administration, their approved sea service will be recalculated from the time of the accident. See: 船员任职期间的海上服务资历有没有有效期？是否所有资历都是认可的？[“Is There a Validity Period for the Approved Sea Service of Crew During Their Tenure? Is All Sea Service Recognized?”], 中华人民共和国海事局 [Maritime Safety Administration of the People’s Republic of China], undated, <https://www.msa.gov.cn/page/knowledgeBase.do?action=content&knowledgeBaseId=29258E3F-969D-4304-9FED-C40F194D58C5>, accessed 4 July 2025.

less common, the number of total registered mariners is likely to increase perpetually regardless of their expired certificates, work status, or retirement.<sup>48</sup>

In 2023, there were 943,954 seafarers registered for international and coastal work, but only 443,800 (47 percent) had approved sea service that year.<sup>49</sup> This figure drops further when considering those with higher qualifications. In the same year, there were 232,839 seafarers holding certificates of competency for international seagoing vessels that were qualified to participate in navigational or engine watch duty, and 174,847 seafarers holding certificates of competency for coastal routes who were qualified to participate in navigational or engine watch.<sup>50</sup>

Figure 4: 2023 Captains, Officers, AB-D/AB-E, and RFPNW / RFPEW<sup>51</sup>

| International |               |                |                                                         |
|---------------|---------------|----------------|---------------------------------------------------------|
| Year          | Captains      | Officers       | AB-Deck/AB-Engine/RFPNW/RFPEW/Electrician <sup>52</sup> |
| 2023          | 17,939        | 82,665         | 132,235                                                 |
| Coastal       |               |                |                                                         |
|               | Captains      | Officers       | AB-Deck/AB-Engine/RFPNW/RFPEW/Electrician               |
| 2023          | 17,918        | 69,502         | 87,427                                                  |
| <b>Total:</b> | <b>35,857</b> | <b>152,167</b> | <b>219,662</b>                                          |

More specific data are available on the overall supply and demand of merchant mariners in China. Shanghai Maritime University's China International Maritime Research Center produces the Chinese Seafarer Supply and Demand Index (CSSDI)(中国海员供求指数) and its accompanied reports based on authoritative government data from the Ministry of Transport and Maritime Safety Administration (MSA), as well as survey and census data from maritime academies and shipping companies.<sup>53</sup> This index provides greater fidelity in the exact distribution of seafarer labor in the Chinese market, including merchant mariner grades, certificate holders, vessel classes, and other parameters, focusing on coastal and unlimited navigational zones.

<sup>48</sup> 姚文兵 [Yao Wenbing], 中国海员队伍发展现状分析 [“Analysis on the Current Situation of China's Seafarer Development”], 世界海运 [World Shipping], No. 10, 2020, <https://www.xindemarinenews.com/topic/chuanrenxinsheng/2020/0924/23851.html>.

<sup>49</sup> “2023 Annual Report on Chinese Seafarer Development,” p. 3.

<sup>50</sup> Ibid, pp. 4-5.

<sup>51</sup> Ibid.

<sup>52</sup> This refers to ratings that are qualified to perform watchkeeping duties as part of a navigational or engineering watch. This could include wheel/anchor/gangway watch, taking rudder orders, keeping a lookout, and assisting with other tasks under the watch command. They are referred to in Chinese as 高级值班水手, 高级值班机工, 值班水手, 值班机工, and 电子技工. They do not include the ordinary seaman, able-bodied seaman, and other ratings that do not participate in watch standing duties.

<sup>53</sup> 中国海员供求指数介绍 [“Introduction to the Chinese Seafarer Supply and Demand Index”], 国际海事(中国)研究中心 [China International Maritime Research Center], Undated, <https://cimrc.shmtu.edu.cn/2020/1106/c5087a30653/page.htm>, accessed 6 September 2025; 中国海员供求指数发布暨研究报告-2021年版 [“Release and Research Report on the Chinese Seafarer Supply and Demand Index – 2021 Edition”], 国际海事(中国)研究中心 [China International Maritime Research Center], 2021, p. 19, <https://cimrc.shmtu.edu.cn/2021/1008/c5087a136635/page.htm>.

The mid-to-late 2010s saw an oversupply in Chinese seafarers in unlimited navigational zones and some shortages in coastal. The 2019 index report provided longitudinal data to highlight the growing disparity between the number of registered seafarers and those that actually worked on board vessels. From 2014-2018, the number of registered seafarers of all ranks in unlimited navigation areas grew from 447,054 to 545,877, while those that recorded sea time onboard vessels only grew from 254,555 to 258,246. This means that 56.94 percent of seafarers worked aboard vessels in 2014, compared to only 47.31 percent in 2018. The figures worsen to 50.26 percent and 42.01 percent respectively when narrowing it down to those who recorded more than three months of sea time. According to the report, this resulted from an oversupply of seafarers across all ranks, but particularly in ratings and some officer positions. Conversely, supply and demand for coastal zone seafarers during this same period experienced dramatic fluctuations and was considered to be in an overall shortage by 2018.<sup>54</sup> These dual trends of severe oversupply in unlimited navigation zones and shortages in coastal navigational zones would continue in 2019.<sup>55</sup>

The oversupply situation changed significantly in 2020, primarily due to the COVID-19 pandemic and its negative impact on global shipping. Unlimited navigation seafarers experienced a clear shift toward demand outpacing supply, with intensifying shortages in third mates, fourth engineers, and some ratings. This was likely due to pandemic restrictions on international crew relief and large numbers of unlimited navigation seafarers transferring to domestic coastal routes. Coastal zone seafarer labor supply continued to experience both oversupply and shortages: that is, there were severe shortages in some positions, such as third officer, fourth engineer, motorman, and captain, with severe oversupply of second officers and second engineers. Overall, the index considered coastal zone labor supply to be in general undersupply. Those actively working aboard coastal navigation vessels grew to over 65 percent while unlimited navigation seafarers actively logging sea time dropped to just 32.11 percent.<sup>56</sup>

According to the overall assessment of the 2021 CSSDI report, China is experiencing a general reduction in the number of seafarers that could result in an overall seafarer shortage (全面性海员紧缺).<sup>57</sup> China is not alone in this situation as the global seafarer labor market has experienced critical shortages. As one of the primary nations supplying global seafarers, these international and domestic labor trends are interrelated and acutely felt in China.<sup>58</sup>

Assessments of this trend note that the merchant marine career has lost the broad appeal in China that it once enjoyed, exacerbated by an aging workforce. This trend can be seen in the difference between enrollment numbers in maritime schools and those who actually passed certificate examinations to enter the workforce upon graduation. Enrollment typically hovers around 15,000 nationally, but only

---

<sup>54</sup> 中国海员供求指数发布暨研究报告- 2019 年版 [“Release and Research Report on the Chinese Seafarer Supply and Demand Index – 2019 Edition”], 国际海事(中国)研究中心 [China International Maritime Research Center], 2019, pp. 6-7, 9, [www.sh.msa.gov.cn/u/cms/www/202201/131559581642060798873.pdf](http://www.sh.msa.gov.cn/u/cms/www/202201/131559581642060798873.pdf).

<sup>55</sup> 中国海员供求指数发布暨研究报告- 2020 年版 [“Release and Research Report on the Chinese Seafarer Supply and Demand Index – 2020 Edition”], 国际海事(中国)研究中心 [China International Maritime Research Center], 2020, p. 4, [www.sh.msa.gov.cn/u/cms/www/1600937305057.pdf](http://www.sh.msa.gov.cn/u/cms/www/1600937305057.pdf).

<sup>56</sup> “Release and Research Report on the Chinese Seafarer Supply and Demand Index – 2021 Edition,” p. 8.

<sup>57</sup> Ibid, pp. 3-4.

<sup>58</sup> “Shipping and World Trade: Global Supply and Demand for Seafarers,” *International Chamber of Shipping*, Undated, <https://www.ics-shipping.org/shipping-fact/shipping-and-world-trade-global-supply-and-demand-for-seafarers/>, accessed 2 September 2025.

one-third of these graduates actually obtains their third officer/fourth engineer certificates.<sup>59</sup> This suggests that graduates do not consider the merchant marine to be a relatively desirable or competitive career path. Other surveys reinforce this trend. Surveys tracking the rate of Dalian Maritime University graduates with maritime majors choosing to work aboard ships found a strong decline since 2000, with only 50 percent choosing to work at sea within five years after graduating. Many graduates entering the workforce also leave soon after. One COSCO Shipping sampling analysis indicated that the attrition rate of fresh graduates within one year was as high as 30%, and 65% after five years.<sup>60</sup> Poor job retention after graduation could result in serious shortages of future officers as older mariners leave the workforce.

The merchant marine is also losing numbers as some personnel choose not to return to work. This trend has gradually worsened with younger generations that join the merchant marine and find less fulfillment in the career and less attraction to the pay.<sup>61</sup> Analysis and discussion of this in China covers a broad range of reasons for personnel weighing whether to leave the merchant marine. These include the hardships of shipboard work and life, such as isolation, diet, and the prison-like environment of the ship. The vessel living conditions of major shipping companies are generally tolerable, but conditions aboard many coastal route vessels are often considered deplorable. The labor marketplace also contains many intermediary agencies with bad practices, sometimes resulting in delays in salary payout and other issues. Atop all these conditions, work at sea is inherently dangerous, naturally causing many seafarers to carefully consider whether work conditions are worth the personal risk. This risk is worsened by safety concerns over reduced crewing and overworked crews.<sup>62</sup>

Some issues are socially driven. Seafarers no longer enjoy the prestige they once did. Senior officers are sometimes not treated with appropriate authority by junior personnel and society has a negative impression of the profession.<sup>63</sup> To address this, China has sought to revamp promotion of the profession nationally. This was one of the goals of the Seafarer Development Plan, which involved propaganda campaigns and the introduction of policies favorable to the profession and its competitiveness. Every year, China uses major dates such as the IMO's Day of the Seafarer (世界海员日) and the National Seafarer Skills Competition (全国海员技能大比武) on June 25<sup>th</sup>, China

---

<sup>59</sup> Yao, "Analysis on the Current Situation of China's Seafarer Development"; 钱华忠 [Qian Huazhong], 后疫情时代海员队伍建设探讨 ["Discussion on Building the Merchant Marine in the Post-Pandemic Era"], 信德海事网 [Xinde Marine News], 13 November 2021, <https://www.xindemarineneeds.com/topic/chuanrenxinsheng/2021/1113/34031.html>.

<sup>60</sup> 全球性海员短缺, 如何吸引并留住海员? ["There is a Global Shortage of Seafarers, How Do We Attract and Retain Them?"], 中国海事网 [China Maritime Safety Online], 13 August 2019, <https://www.zgsyb.com/news.html?aid=512093>.

<sup>61</sup> Numerous personal experiences are shared online. Some can be seen in social media or in professional publications. 关于中国海员存在问题的讨论 ["Discussion on Problems With Chinese Seafarers"], 海运圈专栏作者: 航海衣羊(胡月祥船长) [HYQFocus Columnist: Hang Hai Yi Yang (Captain Hu Yuexiang)], 17 October 2016, <https://www.hyqfocus.com/hyqfocus/jsp/model.jsp?id=133&modelType=2>; 游懿 [You Yi], 高素质船员队伍建设路径探讨 ["Path Exploration of Building a High-Quality Seafarer Team"], 中国海事 [China Maritime Safety], No. 7, 2024, p. 11.

<sup>62</sup> 蔡文佳 [Cai Wenjia], 深度: 我国船员流失, 根源在这儿.. ["In Depth: The Root Cause of the Loss of China's Seafarers Lies Here ..."], 珠江海运 [Zhujiang Shipping], 5 August 2019, <https://www.hyqfocus.com/jsp/model.jsp?id=14637&modelType=1>; Qian Huazhong, "Discussion on Building the Merchant Marine in the Post-Pandemic Era."

<sup>63</sup> Ibid; 蔡文佳 [Cai Wenjia], 深度: 我国船员流失, 根源在这儿.. ["In Depth: The Root Cause of the Loss of China's Seafarers Lies Here ..."], 珠江海运 [Zhujiang Shipping], 5 August 2019, <https://www.hyqfocus.com/jsp/model.jsp?id=14637&modelType=1>.

National Maritime Day (中国航海日) on July 11<sup>th</sup>, along with provincial initiatives, to promote seafarer career development.<sup>64</sup>

Merchant marine pay is not as competitive compared to shore-based jobs as it once was. China does not maintain official data on seafarer pay, but it can be found in industry survey data. One assessment found that pay significantly declined for operations-grade personnel from 2010-2017, likely due to seafarer oversupply.<sup>65</sup> The following few years saw stagnating or declining pay for most ship positions, except on oceanic oil and chemical tankers. Compared to bulk or container markets, oil and chemical industries are somewhat more stable and require more specialized certifications for seafarers, resulting in steady pay increases over time. Bulk cargo lines saw the lowest increases in crew pay, followed by containership positions.<sup>66</sup> These worsening trends of reduced prestige, under-competitiveness in pay, and a declining rate of workforce entry were highlighted in the 2017 *Chinese Seafarer Development Report*, which concluded that it will be difficult for the workforce to meet the needs of the PRC's shipping development.<sup>67</sup>

The COVID-19 Pandemic wrought massive disruption to the global seafarer market. During the pandemic's height, hundreds of thousands of seafarers were stranded and unable to get relief and repatriation due to travel restrictions. Many stranded seafarers experienced fatigue, fear, and mental health problems during this period, stuck at sea and unable to go ashore. Those who were able to get relieved still had problems returning home and faced several weeks of quarantine, with newly mutating strains of COVID-19 causing even further complications.<sup>68</sup> Restrictions were particularly strict in China, which implemented a zero-COVID policy that entailed mass testing, mandatory quarantine, and limits on movement.<sup>69</sup>

---

<sup>64</sup> "Chinese Seafarer Development Plan (2016-2020)"; 2025 年中国航海日公告 ["Notification on the 2025 China Maritime Day"], 新华社海口 [*Xinhua Haikou*], 11 July 2025, [https://www.gov.cn/yaowen/liebiao/202507/content\\_7031669.htm](https://www.gov.cn/yaowen/liebiao/202507/content_7031669.htm).

<sup>65</sup> 中国海员队伍发展现状、问题与对策 ["The Current Status, Issues, and Countermeasures of Chinese Seafarers"], 船管网 [*Ship Management*], 11 November 2017, <http://www.shipmg.com/html/281.html#>.

<sup>66</sup> 船员工资十年变化大盘点：三副三管十年薪资翻 3 倍? ["A Look Back at Changes in Seafarer Wages Over the Past Decade: Have the Salaries of Third Officers and Third Engineers Tripled in the Past Decade?"], 航运 e 家 [*hyej.net*], 15 August 2025, [https://www.hyej.net/cms/detail\\_543.html](https://www.hyej.net/cms/detail_543.html).

<sup>67</sup> "2017 Annual Report on Chinese Seafarer Development," p. 57.

<sup>68</sup> The IMO created a Seafarer Crisis Action Team to address individual cases and provide assistance to seafarers in distress. See: "Seafarer Crisis Action Team: Case Studies," *International Maritime Organization*, Undated, <https://www.imo.org/en/mediacentre/hottopics/pages/scat.aspx>, accessed 5 September 2025; Jackie Northam, "Many Seafarers Are Stranded Aboard Ships Because of the Corona Virus," *National Public Radio – Morning Edition*, 28 January 2021, <https://www.npr.org/2021/01/28/961470026/many-seafarers-are-stranded-aboard-ships-because-of-the-coronavirus>.

<sup>69</sup> It covered many specific measures of COVID-19 prevention, including the issuance of personal protective equipment, testing regimens, crew changes and quarantine procedures, seafarer mental health treatment, daily health reporting, and much more. At least twelve versions of this guidance were published. See: 船舶船员新冠肺炎疫情防控操作指南(第十二版) ["The Operational Guidelines on the Prevention and Control of COVID-19 for Seafarers (V. 12)"], 中华人民共和国海事局 [*Maritime Safety Administration of the People's Republic of China*], 12 December 2022, <https://xindemarinews.com/data/haishifagui/2022/1214/44205.html>; 关于新冠肺炎疫情防控期间国际航行船舶中国籍船员换班操作要求 ["The Operational Requirements on Prevention and Control for International Route Ships Conducting Chinese Crew Change During COVID-19"], 国务院应对新型冠状病毒肺炎疫情联防联控机制综合组 [*Joint Prevention and Control Mechanism of the State Council in Response to the Novel Coronavirus Pneumonia*], 10 December 2021, [https://cdc.fuzhou.gov.cn/zz/jkdt/jkyw/202112/t20211228\\_4279467.htm](https://cdc.fuzhou.gov.cn/zz/jkdt/jkyw/202112/t20211228_4279467.htm); In many cases, local authorities

The significant difficulty experienced during this time sparked temporary increases in seafarer pay as companies needed their workers to endure these hardships to keep ships in operation. Third officers working on major international container lines earning average monthly pay of 1,900 USD in 2019 saw their monthly wages grow to 4,700 USD in 2021. All positions in various vessel classes received increases. Captains enjoyed an increase of around 1,400 to 2,000 USD per month.<sup>70</sup> On 1 January 2022, China Implemented new pay standards for seafarers during the pandemic. Established through industry consultation, a consensus was reached that Chinese oceangoing seafarers should receive a 7 percent increase in minimum base salary for the 2022-2023 year based on 2021 levels.<sup>71</sup> Some in China viewed its strict pandemic control measures as a boon to the competitive advantage of Chinese seafaring labor, claiming it helped keep ships operational through the tumultuous period.<sup>72</sup>

After China ended its zero-COVID-19 rules in January 2023, PRC seafarer wages gradually receded from pandemic peaks as market forces recalibrated.<sup>73</sup> Nonetheless, still-elevated PRC seafarer wages have led shipowners to reduce pay or seek out cheaper foreign labor for “mixed ship manning” (混派). According to one PRC ship captain writing from a Chilean port in July 2023, this trend began in earnest after the impact of COVID discouraged Chinese seafarers from the profession and the large increases in seafarer pay. He further noted that most private shipowners in China are using foreign crewing from the Philippines, India, Myanmar, Bangladesh, Pakistan, Vietnam, and other Asian countries.<sup>74</sup>

---

prohibited PRC vessels from conducting crew changes in domestic ports. Zhiwei Zhao, Li Juntang, and Yueyan Wu, “Fatigue During the COVID-19 Pandemic: The Experiences of Chinese Seafarers,” *Marine Policy*, 153 (2023), p. 2.

<sup>70</sup> “A Look Back at Changes in Seafarer Wages Over the Past Decade: Have the Salaries of Third Officer and Third Engineers Tripled in the Past Decade?”

<sup>71</sup> 疫情期间船员劳动待遇标准出台 [“Labor Pay Standards For Crew Members During the Pandemic are Released”], 新华网 [*Xinhua*], November 22, 2021, [http://www.news.cn/politics/2021-11/22/c\\_1128089179.htm](http://www.news.cn/politics/2021-11/22/c_1128089179.htm).

<sup>72</sup> “A Look Back at Changes in Seafarer Wages Over the Past Decade: Have the Salaries of Third Officers and Third Engineers Tripled in the Past Decade?”; 疫情下中国船员正撑起全球贸易, 被各船公司争抢 [“Chinese Seafarers Are Supporting Global Trade During the Pandemic and Are Being Sought After By Shipping Companies”], 财新网 [*Caixin*], 18 June 2021, <https://www.zgsyb.com/news.html?aid=594299>.

<sup>73</sup> Ibid; 中国海员薪资水平逐渐恢复到疫情前水平 [“Chinese Seafarers’ Wages Are Gradually Returning to Pre-Pandemic Levels”], 信德海事网 [*Xinde Marine News*], 4 September 2023, <https://www.xindemarineneews.com/job/haiyuanxinzi/2023/0904/49888.html>;

<sup>74</sup> 李晓旭 [Li Xiaoxu], 浅谈船员混派感想 [“A Brief Discussion on Mixed Crewing”], 信德海事网 [*Xinde Marine News*], 24 July 2023, <https://www.xindemarineneews.com/topic/chuanrenxinsheng/2023/0724/49137.html>. The trend toward foreign crewing was echoed in other commentaries, suggesting that Chinese seafarers will continue to be competitive in higher ranking positions, while most of the ratings in time might be given to foreign labor. 中国海员工资飙升? 中国海员现状及未来的几点看法 [“Are Chinese Seafarers’ Wages Soaring? Some Perspectives on the Current Situation and Future of Chinese Seafarers”], 信德海事网 [*Xinde Marine News*], 23 July 2024, <https://www.xindemarineneews.com/topic/chuanrenxinsheng/2024/0723/55501.html>.

Figure 5: China (Shanghai) International Seafarers' salary table drawn from 18 Shanghai-based seafarer service and ship management companies.<sup>75</sup>

| China (Shanghai) International Seafarers' Monthly Salary Table<br>(26 August 2025) (US Dollars) |           |            |        |          |
|-------------------------------------------------------------------------------------------------|-----------|------------|--------|----------|
| Position                                                                                        | Container | Bulk Cargo | Tanker | Chemical |
| Captain                                                                                         | 9,587     | 9,421      | 12,111 | 11,184   |
| Chief Officer                                                                                   | 7,612     | 7,137      | 9,255  | 8,818    |
| 2nd Officer                                                                                     | 4,597     | 4,412      | 5,284  | 5,219    |
| 3rd Officer                                                                                     | 4,299     | 4,158      | 4,951  | 4,939    |
| Chief Engineer                                                                                  | 9,237     | 9,014      | 11,354 | 10,430   |
| 2nd Engineer                                                                                    | 7,629     | 7,116      | 9,273  | 8,739    |
| 3rd Engineer                                                                                    | 4,592     | 4,404      | 5,309  | 5,227    |
| 4th Engineer                                                                                    | 4,291     | 4,150      | 4,965  | 4,946    |
| Electrical Engineer                                                                             | 4,816     | 4,417      | 5,452  | 5,214    |
| Boatswain                                                                                       | 2,186     | 2,054      | 2,745  | 2,646    |
| Chief Motorman                                                                                  | 2,181     | 2,051      | 2,745  | 2,646    |
| Able-bodied Seaman                                                                              | 1,735     | 1,541      | 1,911  | 1,880    |
| Motorman                                                                                        | 1,735     | 1,541      | 1,911  | 1,880    |
| Cook                                                                                            | 2,122     | 2,024      | 2,459  | 2,294    |
| Waiter                                                                                          | 1,023     | 767        | 1,173  | 1,088    |

### Managing the Seafarer Workforce—the Maritime Safety Administration (MSA)

The MSA, under the Ministry of Transport, is the core government organ for administering China's merchant marine. It is responsible for managing waterway safety, vessel registrations and inspections, prevention of ship pollution, and administering the examination, credentialing, and supervision of the merchant marine.<sup>76</sup> The MSA is primarily an administrative and law enforcement government organ, separate from the China Coast Guard (中国海警), which is a law enforcement organization chiefly responsible for maritime rights protection and law enforcement under the Central Military Commission's military chain of command.<sup>77</sup> This separation of responsibilities relieves the Coast Guard of the heavy burden of administering China's massive merchant fleet and workforce.

<sup>75</sup> 中国(上海)国际海员薪酬指数 ["China Crew's Remuneration Index"], 上海航运交易所 [Shanghai Shipping Exchange], Updated 26 August 2025, <https://www.sse.net.cn/index/singleIndex?indexType=ccri>; 指数简介 ["Index Introduction"], 上海航运交易所 [Shanghai Shipping Exchange], Undated, <https://www.sse.net.cn/indexIntro?indexName=ccri>, accessed 11 September 2025.

<sup>76</sup> "The Main Responsibilities of the Maritime Safety Administration (Ministry of Transport) of the People's Republic of China" [中华人民共和国海事局（交通运输部海事局）主要职责], 中华人民共和国海事局 [Maritime Safety Administration of the People's Republic of China], 25 May 2021, <https://www.msa.gov.cn/page/article.do?articleId=86183E24-A5E6-4476-9DB6-B954A3E287DE>.

<sup>77</sup> 搞不清海事、海关、海警、海监有什么区别的看过来! ["If You Are Confused About the Differences Between the MSA, Customs, Coast Guard, and Maritime Surveillance, Please Take a Look!"], 中国海洋发展研究中心 [Academy of Ocean of China], 5 May 2023, <https://aoc.ouc.edu.cn/2023/0505/c15171a431605/pagem.htm>.

The MSA Headquarters (海事局机构) in Beijing, located within the Ministry of Transport complex, contains 21 subordinate offices responsible for various functions, policymaking, coordination, and supervision of the 15 directly subordinate regional MSA branches across China's maritime provinces and regions.<sup>78</sup>

MSA offices play critical roles in building the national maritime administrative system within which the merchant marine operates. This includes offices that plan waterway and vessel routes, manage marine radio frequencies and usage, vessel registration, ship safety compliance, coastal navigational aids, national maritime emergency response coordination, overseas international coordination, and many other functions to keep waterborne commerce flowing efficiently.<sup>79</sup> These structures are mirrored in the 15 directly subordinate MSA branches.<sup>80</sup>

The Mariner Management Office (船员管理处) is the most critical MSA office for administering the nation's merchant marine. It drafts laws, regulations, and rules related to crew management and supervises their implementation. It develops and supervises the qualification standards for mariners, administers their training, examination, and certification, and manages the qualifications of training institutions engaged in mariner training. It also develops minimum safe manning standards for ships

---

<sup>78</sup> 内设机构 [“Internal Structure”], 中华人民共和国海事局 [Maritime Safety Administration of the People's Republic of China], 7 April 2024, <https://www.msa.gov.cn/page/article.do?articleId=45AF5734-8B5D-429C-BAB3-F8EFD079E1AC&channelId=7B1BBBD5-BBF4-438F-8CDA-B026A8B5003F>.

<sup>79</sup> Ibid.

<sup>80</sup> For examples, see the Jiangsu MSA (江苏海事局), the Guangdong MSA (广东海事局), and the Lianyungang MSA (连云港海事局). 江苏海事局组织机构图 [“Organizational Structure Chart of the Jiangsu Maritime Safety Administration”], 中华人民共和国江苏海事局 [People's Republic of China Jiangsu Maritime Safety Administration], Undated, <https://www.js.msa.gov.cn/col/col59/index.html>, accessed 17 August 2025; 广东海事局 [“Guangdong Maritime Safety Administration”], 中华人民共和国海事局 [People's Republic of China Maritime Safety Administration], 20 February 2020, <https://www.msa.gov.cn/page/article.do?articleId=C87C4C6C-B116-4479-A7FA-FC37F2895626&channelId=531631DD-852A-4B19-86E1-C6029A27ECA3>; 连云港海事局 [“Lianyungang Maritime Safety Administration”], 中华人民共和国海事局 [People's Republic of China Maritime Safety Administration], 20 February 2020, <https://www.msa.gov.cn/page/article.do?articleId=828F3D30-01ED-4446-8BA5-E3D1440E3FB6&channelId=531631DD-852A-4B19-86E1-C6029A27ECA3>; There are also three regional navigational support centers directly subordinate to the Ministry of Transport but included under MSA management, which build and maintain maritime navigational aids, survey and map port channels, and maintain maritime safety communications. The three centers are the Northern, Eastern, and Southern Navigation Support Centers (南海航海保障中心). 北海航海保障中心 [“Northern Navigation Support Center”], 交通运输部海事局 [Maritime Safety Administration of the Ministry of Transport], 20 February 2020, <https://www.msa.gov.cn/page/article.do?articleId=F14A3BCB-0D33-4FA8-B096-F0AC2DF3D906&channelId=531631DD-852A-4B19-86E1-C6029A27ECA3>; 东海航海保障中心 [“Eastern Navigation Support Center”], 交通运输部海事局 [Maritime Safety Administration of the Ministry of Transport], 20 February 2020, <https://www.msa.gov.cn/page/article.do?articleId=27B47C50-30F7-4A5F-B14E-19762BE0AA0A&channelId=531631DD-852A-4B19-86E1-C6029A27ECA3>; 南海航海保障中心 [“Southern Navigation Support Center”], 交通运输部海事局 [Maritime Safety Administration of the Ministry of Transport], 20 February 2020, <https://www.msa.gov.cn/page/article.do?articleId=D56F5F3D-8684-4DC3-BC19-AD1266C5D7FE&channelId=531631DD-852A-4B19-86E1-C6029A27ECA3>.

<sup>80</sup> “Internal Structure.”

<sup>80</sup> For an example, see the Hainan MSA organization structure. 机构设置 [“Structure”], 中华人民共和国南海海事局 [Hainan Maritime Safety Administration of P.R.C.], Undated, [https://hn.msa.gov.cn/hszc\\_2\\_2/index.jhtml](https://hn.msa.gov.cn/hszc_2_2/index.jhtml).

<sup>80</sup> 船员考试中心 [“Mariner Testing Center”], 中华人民共和国南海海事局 [Hainan Maritime Safety Administration of P.R.C.], 28 April 2024, [https://hn.msa.gov.cn/hszc\\_2\\_2\\_3/18640.jhtml](https://hn.msa.gov.cn/hszc_2_2_3/18640.jhtml).

<sup>80</sup> 中华人民共和国海事局 – 海事一网通办 [“The People's Republic of China Maritime Safety Administration – MSA One-Stop Service”], Undated, <https://zwfw.msa.gov.cn/>. Accessed 20 August 2025.

and manages manning certificates (配员证书).<sup>81</sup> This headquarters-level office oversees the Mariner Management Offices in all the directly-subordinate MSAs, which in turn oversees the Mariner Management Offices in the local MSAs and their subordinate branches.<sup>82</sup>

Mariner Testing Offices (船员考试中心) implement the examination system, including issuing training certificates, maintaining testing facilities and computer systems, ensuring the integrity of the exams, grading those exams, and assisting the Mariner Management Offices in the daily supervision and management of mariner education and training institutions within their jurisdiction.<sup>83</sup>

Merchant mariners, or their employers, can access numerous services on the official MSA website. In the MSA's online integrated information service, called "MSA One-Stop Service" (海事一网通办), mariners can look up test scores, exam schedules, find medical examination locations, check crew service agencies, view vessel information and inspection results, and access various other services and information. They can submit forms and apply for their merchant mariner documents using this online system or go to a service window at a nearby MSA branch.<sup>84</sup>

### **Merchant Marine Training, Examination, and Credentialing**

The Ministry of Transport promulgates the "Rules for the Examination and Licensing of Seafarers of the People's Republic of China" (中华人民共和国海船船员适任考试和发证规则) (referred to in shorthand as "The 20 Rules" / "20 规则"), which stipulates the rules governing merchant marine licensing and qualifications. It is periodically updated to reflect industry and regulatory imperatives, with the most recent version updated in July 2020 and amended in April 2022.<sup>85</sup> This section will explore this credentialing system in greater detail as all PRC seafarers live, work, and progress in their careers according to these rules. Much like the rest of the world's merchant marine, Chinese seafarers navigate a complex system of certification based on their qualifications, training, sea time, and the classes of vessels they serve aboard.

PRC merchant mariners are licensed according to the navigational areas in which they work and the vessels on which they serve. Navigational zones, as mentioned previously, are divided into unlimited navigation, coastal navigation, and inland waterways. This section will focus exclusively on seafarer licensing for unlimited and coastal navigation zones.<sup>86</sup> Rules for "inland waterways" are covered under separate regulations.<sup>87</sup>

---

<sup>81</sup> "Internal Structure."

<sup>82</sup> For an example, see the Hainan MSA organization structure. "Structure," *Hainan Maritime Safety Administration of P.R.C.*.

<sup>83</sup> 船员考试中心 ["Mariner Testing Center"], 中华人民共和国海南海事局 [*Hainan Maritime Safety Administration of P.R.C.*], 28 April 2024, [https://hn.msa.gov.cn/hszc\\_2\\_2\\_3/18640.jhtml](https://hn.msa.gov.cn/hszc_2_2_3/18640.jhtml). If/when the PRC state wanted to exercise control of PRC-flagged vessel(s) for military use, this office would play a supporting role, but ultimately defense mobilization occurs through the civil-military national defense mobilization committee structure.

<sup>84</sup> "The People's Republic of China Maritime Safety Administration – MSA One-Stop Service."

<sup>85</sup> "Rules for the Examination and Licensing of Seafarers of the People's Republic of China."

<sup>86</sup> Article 63, *Ibid.*

<sup>87</sup> See 《中华人民共和国内河船舶船员适任考试和发证规则》 ["Rules of the People's Republic of China for Competency Examination and Certification of Inland Vessel Crew"], 交通运输部令 2015 年第 21 号 [*Ministry of Transport Order 2015 No. 21*], Amended on 6 July 2020, [https://xxgk.mot.gov.cn/2020/gz/202112/t20211222\\_3632485.html](https://xxgk.mot.gov.cn/2020/gz/202112/t20211222_3632485.html).

Figure 6 (below) breaks down the levels of unlimited navigation and coastal navigation certification for officers and ratings, providing a sense of what size ships and power plants they qualify for.

Figure 6: Certificates of Competency for Seagoing Vessels<sup>88</sup>

| <b>Officer Unlimited Navigation Certificate of Competency (无限航区适任证书)</b>                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| There are two levels of unlimited navigation certificates for captains, chief officers, chief engineers, and second engineers:<br>First-class certificate (一等适任证书): ships of or greater than 3,000 gross tons or 3,000 kw plants;<br>Second-class certificate (二等适任证书): ships of 500-3,000 gross tons or 750-3,000 kw plants.                                                             |
| Second officers, third officers, third engineers, and fourth engineers with unlimited navigation certificates are qualified for ships of or greater than 500 gross tons or 750 kw plants.                                                                                                                                                                                                 |
| <b>Officer Coastal Navigation Certificate of Competency (沿海航区适任证书)</b>                                                                                                                                                                                                                                                                                                                    |
| There are three levels of coastal navigation certification for captains, chief officers, chief engineers, and second engineers:<br>First-class certificate: ships of or greater than 3,000 gross tons or 3,000 kw plants;<br>Second-class certificate: ships of 500-3,000 gross tons or 750-3,000 kw plants;<br>Third-class certificate: ships less than 500 gross tons or 750 kw plants. |
| There are three levels of coastal navigation certification for second officers, third officers, third engineers, and fourth engineers:<br>First-class certificate: ships of or greater than 500 gross tons or 750 kw plants;<br>Second-class certificate: ships less than 500 gross tons or 750 kw plants.                                                                                |
| <b>Certificate of Competency for Ratings (船员适任证书)</b>                                                                                                                                                                                                                                                                                                                                     |
| Certification for AS-D and AS-E allows for ships of or greater than 500 gross tons or 750 kw plants.                                                                                                                                                                                                                                                                                      |
| Certificate levels for RFPNW / RFPEW.<br>Unlimited navigation certificate: ships of or greater than 500 gross tons or 750 kw plants;<br>Coastal navigation certificates are divided into two levels:<br>First-class certificate: ships of or greater than 500 gross tons or 750 kw plants;<br>Second-class certificate: ships less than 500 gross tons or 750 kw plants.                  |
| Ratings that do not participate in a navigation or engine watch are not issued different levels of certification.                                                                                                                                                                                                                                                                         |

To work aboard various classes of vessels and navigational areas, mariners must complete the corresponding training and assessment requirements to receive their qualifications. This includes basic safety training (基本安全培训) and competency training (船员适任培训). Additional training, per separate regulations, is required for special types of vessels, including oil tankers, chemical tankers, liquified natural gas tankers, passenger ships, high-speed ships, ships using gas or other low-flashpoint fuels, or ships transiting polar regions.<sup>89</sup> Typically, a mariner will complete training requirements at training agencies or institutions that offer their specific training programs and, upon completion, receive a “mariner training certificate” (船员培训证明). This document is proof they have completed the training and prerequisite for participation in the corresponding qualification examination.<sup>90</sup>

<sup>88</sup> Article 8, “Regulations on Examination and Certification of Competence for Seafarers of the People's Republic of China.”

<sup>89</sup> Article 24, Ibid.; Article 31 of “Regulation of the People’s Republic of China on Seamen (2023 Revision).”

<sup>90</sup> 船员职业常识百问百答 [“100 Questions and Answers on Seafarer Occupational Knowledge”], 中国海事局 [China MSA], June 2022, pp. 6-7, [www.msa.gov.cn/public/documents/document/mtex/ndi1/~edisp/20220624111425187.pdf](http://www.msa.gov.cn/public/documents/document/mtex/ndi1/~edisp/20220624111425187.pdf).

As in other countries' merchant marine, to rise through the ranks, a mariner will have to satisfy the requirements for a raise in grade (职务晋升). There are separate career tracks for the deck and engine departments, each requiring similar basic safety and professional skills proficiency training and different job-specific proficiency training (岗位适任培训). An extensive detailing of specific requirements in training, course hours, examination, and expectations is contained in the "Seagoing Vessel Mariner Training Outline" (海船船员培训大纲), published by the Ministry of Transport. This 643-page document covers training requirements for every ship position and vessel type-specific certificate of competency.<sup>91</sup>

There are also several modifications that can be made to a mariner's license, termed in the United States "an increase in scope." This entails the removal of limitations to an existing license, such as expanding the navigational area (航区扩大) and increasing tonnage (吨位提高) or horsepower (功率提高) limits of a license.<sup>92</sup> Mariners will need to satisfy time in position, basic and job-specific training, and English exam requirements; take the relevant license modification exam; and satisfy additional training requirements across different subjects depending on the modification for which they are applying (tonnage or navigational area).<sup>93</sup> Seafarers applying to work on international routes must pass the "Professional English Test for Seafarers on International Voyages" and achieve a 60 percent score on the listening and oral portions.<sup>94</sup> See the Appendix for a description of application requirements for training, qualifications, and examinations for seafarer certificates of competency.

The PRC's seafarer training programs are largely in alignment with the latest standards set forth in the International Convention on Standards of Training, Certification, and Watchkeeping for Seafarers (STCW), for which the PRC became a member state upon the convention's entry into force on 28 April 1984.<sup>95</sup> STCW seeks to set standards in competency and certification for seafarers

---

<sup>91</sup> 海船船员培训大纲 (2021 版) ["Seagoing Vessel Mariner Training Outline (2021 Version)"], 中华人民共和国交通运输部 [People's Republic of China Ministry of Transport], 19 August 2021, [https://www.gov.cn/zhengce/zhengceku/2021-10/12/content\\_5642026.htm](https://www.gov.cn/zhengce/zhengceku/2021-10/12/content_5642026.htm)

<sup>92</sup> 值班水手 航区扩大 ["Expansion of Navigational Area for RFPNW"], 中华人民共和国海事局 [Maritime Safety Administration of the People's Republic of China], Undated, [https://www.msa.gov.cn/page/outer/zxcx/cyzygh/cyjs/hc\\_jiashi/lt500/shuishou\\_navigating.html](https://www.msa.gov.cn/page/outer/zxcx/cyzygh/cyjs/hc_jiashi/lt500/shuishou_navigating.html), accessed 29 August 2025.

<sup>93</sup> For one example, see the official requirements for an increase in tonnage or navigational area for third and second officers here: 三副、二副 航区扩大/吨位提高 ["Navigational Expansion / Tonnage Increase for Third and Second Officers"], 中华人民共和国海事局 [Maritime Safety Administration of the People's Republic of China], Undated, [https://www.msa.gov.cn/page/outer/zxcx/cyzygh/cyjs/hc\\_jiashi/gteq3000/3fu\\_navigating.html](https://www.msa.gov.cn/page/outer/zxcx/cyzygh/cyjs/hc_jiashi/gteq3000/3fu_navigating.html), accessed 29 August 2025.

<sup>94</sup> Seafarers working on coastal routes are exempt from the Maritime English examination requirement. Article 5, Regulation of the People's Republic of China on Seamen (2023 Revision); Articles 3, 6, and 7 of 《国际航行船舶船员专业英语考试和发证办法》 ["Professional English Examination and Certification Measures for Mariners on International Voyages"], 交通部海事局 [Ministry of Transport Maritime Safety Administration], 4 November 2008. For English exam exemptions, review the raise in grade requirements contained on the MSA Career Plan page for seafarers. 船员晋升 – 海船船员驾驶 ["Elevation in Mariner Grades – Navigational Seafarers"], 中华人民共和国海事局 [Maritime Safety Administration of the People's Republic of China], Undated, [https://www.msa.gov.cn/page/outer/zxcx/cyzygh/cyjs/hc\\_jiashi.html](https://www.msa.gov.cn/page/outer/zxcx/cyzygh/cyjs/hc_jiashi.html), accessed 29 August 2025.

<sup>95</sup> The 1978 STCW convention was the first to establish basic requirements on training, certification and watchkeeping for seafarers internationally, which were previously established by individual governments without reference to other nations' practices. The convention was amended in 1995 and 2010, and the PRC has revised its national maritime laws many times over the years to remain in compliance with the latest standards of the convention. 中国加入有效的国际海事公约一览 ["A List of International Maritime Conventions in Effect That China Has Acceded"], 国际航运法律研究 [Research on International Shipping Law], 23 September 2019, <https://xindemarineneews.com/data/haishifagui/2019/0923/15179.html>;

internationally. As a major supplier of seafarers, STCW standards have a significant impact on China's maritime labor market, its maritime education and training system, and the relevant work of the MSA. The MSA's International Cooperation Office is tasked with managing compliance mechanisms and tracking trends in international maritime conventions to ensure the PRC's overall compliance.<sup>96</sup> The PRC's efforts to create, revise, and improve national maritime regulations and laws and oversee their implementation to maintain compliance with IMO instruments is a natural result of China's export-oriented economy, expansive sea trade, and its goal of becoming a shipping power. China's effort to build a world-class merchant marine would necessarily require it to be an internationally compliant one.

### *Joining the Merchant Marine*

Aspiring seafarers can begin their careers from the bottom as one of the ratings or attend one of the many maritime schools to become a licensed officer. Some rise through the ranks without attending a school, referred to in the United States as a "hawsepiper" or in China simply as those that did not attend the academy.<sup>97</sup>

To join as a rating on coastal and international routes, one must be between the ages of 18 and 60, pass the physical health exam, complete basic safety training and pass the corresponding exam, and complete any required professional skills training.<sup>98</sup> Basic safety training is often referred to as the "Four Certificates" (四小证), a safety training course that all seafarers must pass, including four training sub-courses: personal survival, fire prevention and fighting, elementary first aid, and personal safety and social responsibility.<sup>99</sup> After completing these requirements, seafarers receive a "seafarer certificate of competency" (海船船员适任证书) and a "Seaman's Identity Document" (船员服务簿), within which their sea time will be recorded in this book and used for license upgrades. They may then serve in unlimited and coastal navigational areas. Additional training, service time, and good performance and safety record are required to qualify for a RFPNW.<sup>100</sup>

Ratings applying to become third officer or fourth engineer must complete no less than 18 months of sea time, complete the required job-specific training, and then pass the examination. Additionally, they should have performed bridge or engine room watch duties for no less than six months on a vessel of corresponding class under the guidance of the captain or a qualified officer as part of their on-board training (船上见习). Job-specific training courses, as determined by the MSA, include subjects on navigation, ship maneuvering and collision avoidance, ship management, ship structure and cargo transport, and maritime English. They will also need to complete practical assessments in

---

"International Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW)," *International Maritime Organization*, Undated, <https://www.imo.org/en/ourwork/humanelement/pages/stcw-conv-link.aspx>.

<sup>96</sup> "Internal Structure."

<sup>97</sup> A "hawspipe" is the reinforced pipe running through the hull on the bow of a ship, through which the anchor chain will pass. A "hawsepiper" is an informal term in the U.S. Merchant Marine denoting a licensed officer who did not attend a traditional maritime college to receive their license and instead began their career as an unlicensed seaman.

<sup>98</sup> Physical examinations are conducted according to the 国家标准 30035-2021 - 船员健康检查要求 ["National Standard 30035-2021 - Criteria of Physical Examination for Seafarers"], 国家市场监督管理总局, 中国国家标准化管理委员会 [State Administration of Market Regulation / Standardization Administration of China], 2021, <https://openstd.samr.gov.cn/bzgk/gb/newGbInfo?hcno=4AAF38BD344260F1CBBC9D0BF21645BB>.

<sup>99</sup> 船员“四小证”的那些事 ["Things About the 'Four Certificates' for Seafarers"], 海员之家 [[www.54seaman.com](http://www.54seaman.com)], 15 November 2022, [https://www.54seaman.com/news/detail\\_139275.html](https://www.54seaman.com/news/detail_139275.html).

<sup>100</sup> 船员从业及维权指南 ["Guide to Mariner Employment and Rights Protection"], 中国海事局 [*China Maritime Safety Administration*], June 2022, p. 4.

electronic navigational chart display and information systems (ECDIS), route planning, radar, cargo stowage, use of navigational instruments, maritime English, and ship maneuvering, collision avoidance, and bridge resource management. If they are applying for a third officer's license with unlimited navigational area, they must have at least six months of sea time aboard a vessel operating in unlimited navigational areas.<sup>101</sup>

Ascending the ranks from a rating to captain can theoretically be done in eight to ten years given no wasted time and smooth promotions. In reality, it usually takes much longer than ten years.<sup>102</sup> Seafarers must meet sea time requirements within proper vessel class, tonnage, and navigation areas, and then complete the administrative processes of training, examinations, and other steps to ensure they achieve their promotions. In many cases, individuals will experience delays, mistakes, or lack of opportunity depending on market conditions, requiring well over a decade to achieve the rank of captain.<sup>103</sup>

The more common route to becoming a seafarer officer is through the PRC's various maritime academies and institutions. Generally, high school graduates will enroll in these institutions as maritime majors (航海类专业). Eligibility to participate in the seafarer competency examination requires at least two years of full-time maritime secondary vocational education or above; or, for those who completed full-time non-maritime education at a junior college or above, completion of at least 12 months of training for the positions of third officer or fourth engineer. Upon completion of their courses, candidates may apply to take the unlimited navigation third officer / fourth engineer exams. A select few universities offering excellent undergraduate degrees in maritime majors are approved by the MSA to allow their graduates to apply for second officer / second engineer qualification examinations.<sup>104</sup>

### *Academies and Professional Mariner Training Institutions*

The Ministry of Transport and the MSA coordinate the planning of training and education nationally.<sup>105</sup> They administer maritime colleges (航海院校) and training institutions (培训机构)

---

<sup>101</sup> 值班水手/高级值班水手 晋升 三副 [“RFPNW / AB-D Raise in grade to Third Officer”], 中华人民共和国海事局 [Maritime Safety Administration of the People's Republic of China], Undated, [https://www.msa.gov.cn/page/outter/zxcx/cyzygh/cyjs/hc\\_jiashi/gteq500/shuishou\\_3fu.html](https://www.msa.gov.cn/page/outter/zxcx/cyzygh/cyjs/hc_jiashi/gteq500/shuishou_3fu.html), accessed 30 August 2025.

<sup>102</sup> For a career path illustration with excellent detail, see the graphic titled “‘20’ Rules Seafarer Promotion Chart” (“20” 规则海船船员晋升图) in: 水手晋升三副最新规则, 超详细船员晋升攻略 [“The Latest Rules for Seafarers to Be Promoted to Third Officers, Super Detailed Seafarer Promotion Strategy”], 航运 e 家 [hvej.net], 14 March 2022, [https://www.hvej.net/cms/detail\\_91.html](https://www.hvej.net/cms/detail_91.html). There are numerous personal career experiences shared on the Chinese short-form video sharing platform Douyin. For example, a search for 海员最快要多久能考上船长 (“How long does it take for a seafarer to pass the examination to become a captain?”) will result in numerous seafarers sharing the theoretical eight to ten year process as well as their own experience that took 14 years or longer. [www.douyin.com](https://www.douyin.com), accessed 10 October 2025.

<sup>103</sup> For one example, one third officer had to first work as an able seafarer due to a lack of ship positions and then pay his way into a third officer position aboard a ship. To transition from third officer to second officer, he had to convince a shipowner by offering to perform work as a second officer while only receiving the salary of a third officer. There could be many impediments that may slow down a seafarer's promotion process. 船员从实习生升职到船长需要多长时间呢? 远洋船员告诉你答案 [“How Long Does It Take for a Seafarer to Be Promoted from Trainee to Captain? Oceangoing Seafarers Tell You the Answer”], 知乎 - 布谷鸟的记忆 [Zhihu - Cuckoo's Memories], 16 November 2021, <https://zhuanlan.zhihu.com/p/434020828>; 从水手到船长花了 16 年, 做海员几多收获几多憾事? [“It Took 16 Years to Go From Seaman to Captain, What Does One Gain and Regret in Being a Seafarer?”], 搜狐 - 海员 Talk [Sohu - Seafarer Talk], 7 September 2024, [https://news.sohu.com/a/807041373\\_121123888](https://news.sohu.com/a/807041373_121123888).

<sup>104</sup> “100 Questions and Answers on Seafarer Occupational Knowledge,” p. 20.

<sup>105</sup> “Chinese Seafarer Development Plan (2016-2020).”

through the issuing of training licenses. An institution must comply with MSA requirements to be granted a license by having the requisite facilities.<sup>106</sup>

According to the official MSA listing by province and municipality, there are at least 242 maritime schools and training institutions that the merchant marine can access nationally. This figure appears to be a reduction from 310 entities in 2015, which was described in the 2016 Seafarer Development Plan as a system that was undergoing gradual improvement and had basically satisfied demand for maritime education.<sup>107</sup> There are numerous institutions available nationwide, with some offering more advanced educational programs.

All maritime colleges and training institutions are regulated according to the “Rules of the People’s Republic of China for the Administration of Seamen Training” and the specific requirements outlined extensively in “Implementation Measures for the Administration of Seamen Training” and its appendices. These documents detail the MSA-required standards of facilities, equipment, programs, staffing, and practices a training institute must comply with to obtain a license and engage in the business of mariner training, examination, and certification.<sup>108</sup>

Maritime colleges refer to ordinary undergraduate colleges (普通本科院校), higher vocational schools (高等职业学校), and secondary vocational schools (中等职业学校) that have been approved by the Ministry of Education and issued training licenses to offer full-time maritime majors for obtaining licenses for third officers, fourth engineers, and electrical engineers. These majors, including navigational technology, marine engineering, and electronic and electrical engineering, aim to train officers that meet the standards of operational-level seafarers.<sup>109</sup> Most officers will begin their education at one of 59 maritime colleges nationwide (as of the end of 2020). These include 17

---

<sup>106</sup> These requirements include having qualified instructors and management staff, a complete system of management and safety protection, a quality control system to meet Ministry of Transport standards, and meet all other associated standards in training outlines, pollution prevention, and vessel safety. Articles 32-34, “Regulation of the People’s Republic of China on Seamen (2023 Revision).”

<sup>107</sup> Chinese Seafarer Development Plan (2016-2020).

<sup>108</sup> For example, appendix four of the implementation measures, “Standards for Training Venues, Facilities and Equipment, and Teaching Staff of Crew Training Institutions” (船员培训机构培训场地、设施设备以及师资配备标准) breaks down the specific program requirements for providing the basic safety training program, competency training programs for all ship positions, professional skills competency training programs, special training programs for specific classes of ships, and teaching staff standards for both seagoing and internal waterway training pipelines. See 附录 4 – 船员培训机构培训场地、设施设备以及师资配备标准 [“Appendix 4: Standards for Training Venues, Facilities and Equipment, and Teaching Staff of Crew Training Institutions”] in the 《〈中华人民共和国船员培训管理规则〉实施办法》 [“Implementation Measures for the ‘Rules of the People’s Republic of China for the Administration of Seamen Training’”]; 《中华人民共和国船员培训管理规则》(2019 修正) [“Rules of the People’s Republic of China for the Administration of Seamen Training (2019 Amendment)”], 交通运输部令 2019 年第 5 号 [Ministry of Transport Order 2019, No. 5], 5 February 2019, <https://www.msa.gov.cn/page/article.do?type=hsfg&articleId=750624778560491297C2EE78D16D0D62>. The latest measures can be found and downloaded from the MSA: 中华人民共和国海事局关于印发《〈中华人民共和国船员培训管理规则〉实施办法》的通知 [“Notice of the Maritime Safety Administration of the People’s Republic of China on Issuing the Implementation Measures for the ‘Rules of the People’s Republic of China for the Administration of Seamen Training’”], 中华人民共和国海事局 [Maritime Safety Administration of the People’s Republic of China], 8 January 2025, <https://www.msa.gov.cn/page/article.do?articleId=9F7B8968-3750-4195-98F5-A282C6C80B05&channelId=0308A180-57AB-4E0E-9236-20FBED32B5EF>.

<sup>109</sup> 姚文兵 [Yao Wenbing], 赵友涛 [Zhao Youtao], 吴桂涛 [Wu Guitao], 邢永恒 [Xing Yongheng], and 段尊雷 [Duan Zunlei], 中国航海教育的现状与挑战 [“The Current Situation and Challenges of China’s Maritime Education”], 大连海事大学高等教育研究院 [Dalian Maritime University Higher Education Research Institute], October 2021, <https://xindemarinenews.com/china/36264.html>.

ordinary undergraduate colleges, 34 higher vocational schools, and 8 secondary vocational schools. Nearly two-thirds of these were founded or established new maritime majors after the turn of the century when enrollments in maritime colleges began to surge until their peak in 2010.<sup>110</sup> The exact number of colleges may change as some institutions gain or lose their accreditations.

Of the 17 undergraduate colleges, four are the most prominent maritime colleges in China. These include Dalian Maritime University, Shanghai Maritime University, Wuhan University of Technology, and Jimei University in Xiamen.<sup>111</sup> These schools have had longstanding seafarer education programs and are major contributors of talent to maritime industries. They received an outstanding grade in the MSA's 2021 Maritime Education and Training Quality Assessment, which evaluates the quality of merchant marine educational and training programs. Having received a certificate of excellence by the MSA, graduates from these schools are able to apply directly for the second officer, second engineer, and other advanced qualification examinations, giving more students the chance to rapidly advance to higher positions aboard merchant shipping.<sup>112</sup>

While there are plenty of maritime schools to examine, Dalian Maritime University is worth highlighting since it is generally considered the best maritime school for navigation majors. Its graduates are highly sought after by top shipping companies.<sup>113</sup> Dalian's unique position in China's maritime industries and education resulted in the creation of China's first "Modern Shipping Industry and Education Integrated Community" (现代航运行业产教融合共同体) in 2025. This initiative is a combined effort by COSCO Shipping Dalian Investment Company, Dalian Maritime University, and the Dalian Vocational and Technical College, meant to broadly enhance Dalian's maritime innovation

---

<sup>110</sup> Ibid. More recent sources state there are 20 schools offering maritime majors nationwide. 2024 年航海技术专业大学排名及评级结果 ["2024 Maritime Technology University Rankings and Ratings"], 中国大学排行榜 [*China University Ranking*], 12 April 2024, <https://www.cnur.com/major/1572.html>.

<sup>111</sup> Dalian Maritime University, Shanghai Maritime University, and Wuhan University of Technology, were formerly directly under the Ministry of Transport. Some schools are still jointly built and supported by the Ministry of Transport. 有哪些学校是原交通部直属的高校? ["Which Schools Were Previously Directly Under the Ministry of Transport?"], 中华人民共和国交通运输部 - 人事教育司 [*Ministry of Transport of the People's Republic of China - Personnel and Education Department*], 24 February 2025, [https://www.mot.gov.cn/liuanyanzixun/202501/t20250122\\_4162934.html](https://www.mot.gov.cn/liuanyanzixun/202501/t20250122_4162934.html); 四大知名航海院校, 选择哪一个比较好? 这个问题也太简单了吧! ["Which of the Four Renowned Maritime Schools is the Best? That's a Simple Question, Isn't It?"], 网易 - 鄂东轮机长 [*Netease Account - Eastern Hubei Chief Engineer*], 17 May 2025, <https://www.163.com/dy/article/JVP9TFE50552TCA5.html>.

<sup>112</sup> 4 大海事院校航海类本科毕业生, 可直接申请二副二管适任考试 ["Graduates of the Four Major Maritime Academies Can Directly Apply for the Second Officer and Third Engineer Qualification Examinations"], 信德海事网 [*Xinde Marine News*], 16 April 2021, <https://www.xindemarinenews.com/china/28775.html>.

<sup>113</sup> The school's prestige consistently attracts such major seafarer employers as COSCO Shipping and its nationwide subsidiaries and has landed it the title "the cradle of navigators." 中国唯一"211"海事大学, 航海类专业全国最强! ["China's Only '211' Maritime University, The Strongest Navigation Majors in the Nation!"], 搜狐 - 阿右聊教育 [*Sohu Account - A You Chats About Education*], 30 July 2018, [https://www.sohu.com/a/244095667\\_100206354](https://www.sohu.com/a/244095667_100206354); "Which of the Four Renowned Maritime Schools is the Best? That's a Simple Question, Isn't It?"; 学校简介 ["School Introduction"], 大连海事大学 [*Dalian Maritime University*], Undated, <https://www.dlmu.edu.cn/wzsy/xxgk/xxjj.htm>, accessed 24 September 2025; COSCO Shipping Group has long sourced many of its seafarers from Dalian Maritime University and its leadership attaches great importance to the talent developed in the school. 高宁 [Gao Ning], 中远海运集团 2026 届航海类专业毕业生招聘宣讲会在我校举办 ["COSCO Shipping Group Holds a Recruitment Seminar for 2026 Graduates of Maritime Majors at Our School"], 大连海事大学 - 校友事务与学生就业处 [*Dalian Maritime University - Alumni Affairs and Student Employment Office*], 5 September 2025, <https://www.dlmu.edu.cn/info/2075/191848.htm>.

and development through deeper cooperation across the government, industry, and education.<sup>114</sup> Such municipal government-sponsored initiatives will likely further boost the prominence of Dalian Maritime University in fostering seafarer talent.<sup>115</sup>

Noteworthy vocational and junior colleges include the Wuhan Institute of Shipbuilding Technology (武汉船舶职业技术学院), the Qingdao Ocean Shipping Mariners College (青岛远洋船员职业学院), and the Jiangsu Maritime Institute (江苏海事职业技术学院).<sup>116</sup> The Qingdao Ocean Shipping Mariners College in particular is a highly targeted institution belonging to China's largest state-owned shipping conglomerate, COSCO Shipping. Located on a brand-new campus established in September 2021, it features advanced facilities and simulators. The school combines a number of in-house educational institutions, including the COSCO Shipping Group Party School, COSCO Shipping University, COSCO Shipping Research Institute, and the Qingdao Ocean Shipping Vocational College, yielding a scale of educational resources commensurate with the size of COSCO's shipping businesses.<sup>117</sup> Serving these concurrent roles, the institution provides the infrastructure for long-term talent development, ultimately funneling that talent into COSCO's many businesses. COSCO subsidiaries, like COSCO Shipping Seafarer Management Co. Ltd, submit training course requirements to be developed at the school, providing talent tailored directly for company's needs.<sup>118</sup> Additionally, the school uniquely provides training and education for COSCO's ship political commissars.<sup>119</sup>

---

<sup>114</sup> 全国首个现代航运行业产教融合共同体在大连成立! ["The Nation's First Modern Shipping Industry-Education Integrated Community is Established in Dalian"], 大连海事 [Dalian MSA], 1 July 2025, <https://www.xindemarinenews.com/m/view.php?aid=60365>.

<sup>115</sup> In June 2025, the Dalian Municipal Government released a programmatic document entitled "Dalian City's Work Plan for Promoting the Construction of an Internationally Renowned Seafarer Education and Training Center and Seafarer Human Resources Allocation Center" (《大连市推进国际知名的海员教育培训中心和海员人力资源配置中心建设工作方案》), also widely referred to as the "dual seafarer center" (海员双中心) plan. This plan seeks to elevate the role of Dalian as a place for domestic and international merchant marine development. Among its goals, the plan prioritizes the development of high-end talent for new vessel types, technologies, navigational regions, and other emerging areas. It also seeks to expand the scale of Chinese international seafarers by supporting seafarer dispatch companies working in Dalian and attract more advanced mariner talent, provide improved seafarer and business services, give Dalian a greater role in international maritime governance, and other efforts to promote Dalian as a base for merchant marine talent. 关爱海员, 大连准备这么做! ["Caring for Seafarers, Dalian is Ready to Do So!"], 搜狐 - 辽宁省 [Sohu.com - Liaoning Province], 25 June 2025, [https://www.sohu.com/a/907863626\\_175033](https://www.sohu.com/a/907863626_175033).

<sup>116</sup> 比较好一点的航海类大专院校, 推荐这三所, 你认同吗? ["Recommendation off These Three Schools as Better Maritime Colleges, Do You Agree?"], 百度-百家号 [Baidu - Baijiahao], 27 June 2025, <https://baijiahao.baidu.com/s?id=1836048857333004951&wfr=spider&for=pc>.

<sup>117</sup> 学院简介 ["College Introduction"], 青岛远洋船员职业学院 [Qingdao Ocean Shipping Mariners College], undated, <https://www.qmc.edu.cn/cyzl/xyjj/index.html>, accessed 20 September 2025; 组织机构 ["Organizational Structure"], 青岛远洋船员职业学院 [Qingdao Ocean Shipping Mariners College], undated, <https://www.qmc.edu.cn/cyzl/xyjj/index.html>, accessed 20 September 2025.

<sup>118</sup> 中远海运船员管理有限公司参与高等职业教育人才培养年度报告 ["Annual Report on COSCO Shipping Seafarer Management Co. Ltd.'s Participation in Higher Vocational Education and Talent Training"], *COSCO Shipping*, 2022, pp. 5-8.

<sup>119</sup> 石庆贺 [Shi Qinghe] and 马云芹 [Ma Yunqin], 国有航运企业船舶政委队伍建设刍议 - 基于中远集团的调研与思考 ["A Preliminary Discussion on the Construction of Ship Political Commissars in State-Owned Shipping Enterprises"], 青岛远洋船员职业学院学报 [Journal of Qingdao Ocean Shipping Mariners College], No. 3, 2015, pp. 32-35; "Annual Report on COSCO Shipping Seafarer Management Co. Ltd.'s Participation in Higher Vocational Education and Talent Training," p. 9.

The attraction of maritime careers for prospective students has vacillated over the past two decades. National student enrollment was quite low in the early 2000s, peaked at over 40,000 in 2010, and was steady through the rest of the 2010s at above 15,000.<sup>120</sup> National initiatives to improve seafarer quality and career prospects have likely had some impact as enrollment has increased to above 20,000 in 2020 and reached over 23,000 in 2023.

Maritime education is an essential component to the development and success of all of China's maritime industries. These schools and training institutions help develop the seafarer workforce across its entire lifespan, from entry-level to senior positions. They deliver new graduates to the industry as operational-level officers, provide training to existing seafarers seeking to upgrade their license, and keep the workforce compliant with the standards established in the IMO's STCW Convention.

### *Licensing for Military Veterans*

Sea time gained in other maritime fields can be used to advance in China's merchant marine or aid in upgrading to different navigational zones or more advanced licenses. According to article 23 of the "Rules of the People's Republic of China of Seafarer Examination and Certification," personnel who have served on inland waterway vessels, ocean-going fishing vessels, or military vessels may apply for corresponding certificates of competency for seafarers. The highest position for which they may apply is chief officer or second engineer, their experience must be of compatible relevant sea service with good performance records, and they must participate in the required training and pass the exams required for the position for which they are applying.<sup>121</sup> Some may be granted exemption from job-specific training and may directly apply for competency certificate exams and, upon completion of a traineeship, apply to be issued such certificates. These include demobilized personnel on military vessels who have received a post-secondary degree in a navigation major or above from a full-time maritime academy, or a post-secondary degree in a related field such as navigation or marine engineering from a military academy. It also includes personnel with service aboard military vessels as sailors or engineers that are applying for RFPNW or RFPEW certificates.<sup>122</sup>

The merchant marine is historically an attractive career transition for veterans. Shipping companies once had to accept discharged military officers when ordered to do so by the government as part of veteran employment programs, but companies have become far more selective since enterprise reform in the 1990s.<sup>123</sup> Nonetheless, the "military-like" work environment, the important but difficult nature of the work, and the high pay relative to work in cities ashore, make the job suited to many transitioning veterans.<sup>124</sup> Previous and current merchant marine development priorities have

---

<sup>120</sup> "2017 Annual Report on Chinese Seafarer Development," p. 37; "The Current Situation and Challenges in China's Maritime Education."

<sup>121</sup> Article 23, "Rules for the Examination and Licensing of Seafarers of the People's Republic of China."

<sup>122</sup> "100 Questions and Answers on Seafarer Occupational Knowledge," pp. 12-13.

<sup>123</sup> Zhao, Shi, and Feng, *The Ship Political Commissar and His Shipmates – Transformation of the Commissar's Role on Chinese Ships*, pp. 25-26.

<sup>124</sup> According to one account, the downsides of joining the merchant marine such as being away from home for months on end pale in comparison to the existence of becoming a Didi driver or delivering takeout and packages for a living, a common profession for today's urban youth which earns very little. The merchant marine offers a solid salary in a life-long profession that is currently in high demand, one in which they have a unique foundation of skill and discipline to excel in. 了解过海员的生活，才懂退役军人为啥这么爱它 ["Only After Understanding the Lives of Seafarers Can you

emphasized increased recruitment of veterans to strengthen the workforce, seeing this group as a core source of talent.<sup>125</sup>

National programs continue to promote this recruiting pipeline, focusing on professional maritime training programs. In 2020, the Ministry of Transport and the Ministry of Veteran's Affairs jointly implemented "Project Wave Break" (浪花计划) to promote mariner training institutions and shipping companies to recruit retired military personnel through "school-enterprise cooperation and targeted training" (校企合用、定向培养), and provide a platform for retired military personnel to obtain professional mariner qualifications and job placement. In 2020, the MSA planned two batches of personnel placement. The first was for 1,900 personnel placed into professional skills training programs across the country and the second was the placement of 1,200 personnel into degree-granting educational programs in conjunction with an overall expansion of enrollment in maritime majors at higher vocational colleges.<sup>126</sup>

Project Wave Break catalyzed numerous initiatives over the next few years across the provinces to promote veterans transitioning into the merchant marine. For example, Guangxi held a graduation ceremony at Beibu Gulf University for its first group of 100 veteran merchant mariners in April 2021, delivering 40 veterans to jobs in international shipping and 60 veterans to jobs in coastal shipping with the Guangxi Hongxiang Shipping Co. Ltd. (广西鸿翔船务有限公司). This rate of job placement was enabled by a strategic cooperation framework agreement on veteran crew training between the company, Beibu Gulf University, Guangxi and local veterans' affairs organs, and the Qinzhou MSA.<sup>127</sup> By October 2024, this arrangement produced 480 veterans from Beibu Gulf University passing their exams and gaining employment with the company.<sup>128</sup> Similar agreements have been struck between local governments, institutions, enterprises, and MSAs across China, ranging from Sanya to Dalian. These programs offer discounted tuition rates for veterans and are cooperating with numerous shipping enterprises.<sup>129</sup>

In return, veterans from these programs are attractive recruits into China's maritime militia forces. From the Guangxi example, an unknown number of veteran participants were also recruited into the

---

Understand Why Veterans Love It So Much"], 退役人才网 [Tuiyirencai], 23 August 2024, <https://www.163.com/dy/article/JA9VL5DJ0538L3WV.html>.

<sup>125</sup> 中华全国总工会关于加强高素质船员队伍建设的指导意见 ["Guiding Opinions of the Chinese Federation of Trade Unions on Strengthening the Construction of a High-Quality Seafarer Workforce"], 交通运输部行政规范性文件 [Ministry of Transport Administrative Normative Documents], Document No. 41, 2021, p. 1.

<sup>126</sup> 交通运输部、退役军人事务部联手打造退役军人上船就业“直通车” ["The Ministry of Transport and the Ministry of Veteran's Affairs Jointly Created a 'Direct Track' for Veteran Employment on Ships"], 央视网 [CCTV], 5 August 2020, [https://www.mva.gov.cn/xinwen/mtbd/202008/t20200805\\_41490.html](https://www.mva.gov.cn/xinwen/mtbd/202008/t20200805_41490.html).

<sup>127</sup> 刘德安 [Liu De'an] and 蓝陆遥 [Lan Luyao], 浪花计划“拓宽退役军人就业渠道 - 广西首期退役军人船员培训班学员结业走上工作岗位” ["'Project Wave Break' Expands Employment Channels For Veterans – Guangxi's First Batch of Veteran's Crew Training Courses Graduates and Enter Employment"], 中国军网 [China Military Online], 29 April 2021, [http://www.81.cn/gfbmap/content/2021-04/29/content\\_288372.htm](http://www.81.cn/gfbmap/content/2021-04/29/content_288372.htm).

<sup>128</sup> 《广西“浪花计划”退役军人船员培训开班，助力海洋强国建设》 ["Guangxi's 'Project Wave Break' Veterans' Crew Training Begins to Help Build a Maritime Power"], 船员帮 [chuanyuanbang.com], 26 October 2024, <https://chuanyuanbang.com/News/news/id/2783>.

<sup>129</sup> 第一批面向退役军人船员培训招生报名信息 ["Enrollment Information for the First Batch of Veteran Mariner Training"], 中国海事交通运输部海事局官方澎湃号 [Official The Paper Account of the Maritime Safety Administration of the Transport Ministry of China], 5 August 2020, [https://m.thepaper.cn/newsDetail\\_forward\\_8586925](https://m.thepaper.cn/newsDetail_forward_8586925).

Guangxi Hongxiang Shipping Co.’s maritime militia force.<sup>130</sup> In Fujian Province, the Quanzhou Ocean Institute (泉州海洋职业学院) began implementing Project Wave Break in October 2020 and is a prominent military-civilian mariner training institute.<sup>131</sup> Working with the city’s People’s Armed Forces Department (PAFD), this institution established a maritime militia training base in 2016 to provide education and training to local maritime militia forces. By October 2023, the base had trained over 5,000 militia personnel, in addition to providing skills training, political education, and job placement assistance to thousands of veterans.<sup>132</sup> It is unclear if similar arrangements are being implemented elsewhere and what the specific obligations militia recruitment entails.

### *Fraud and Abuse*

In practice, the qualification and certification process for mariners involves many cases of fraud and abuse. Some mariners seeking to ascend the ranks for better positions on vessels or within companies may undertake illegal practices in falsifying time served at sea in their Seaman’s Identity Document. Other issues involve cheating on exams, sharing certificates for personal profit, scams from fake seafarer services, company’s attempting to skirt minimum manning requirements, and many other issues that not just break the law, but also create hidden maritime safety dangers in shipping. These are longstanding issues and the MSA has perpetually sought to crack down on bad actors and promote the industry and seafarer profession.<sup>133</sup>

Provincial MSA branches publicly discuss their efforts at combating fraud in merchant marine reported sea time, qualifications, and other issues. According to Guangxi-based MSA experts, some of the common falsification methods include fake endorsements in Seafarer Identity Documents and falsifying crew member information in the ship’s port entry and departure reporting system. Captains might record false sea service for crew members, often due to company requests, “favors” (人情), or for personal profit, regardless of whether they were present aboard the vessel. While also falsifying

---

<sup>130</sup> 刘德安 [Liu De’an] and 蓝陆遥 [Lan Luyao], “浪花计划”拓宽退役军人就业渠道广西首期退役军人船员培训班学员结业走上工作岗位 [“‘Project Wave Break’ Expands Employment Channels for Veterans. Students from Guangxi’s First Veterans’ Crew Training Program Graduate and Begin Working”], 中国国防报 [*China Defense News*], 29 April 2021, p. 1.

<sup>131</sup> In addition to educating merchant mariners, the Quanzhou Ocean Institute has been recognized by authorities as an advanced institution for recruiting PLA NCOs, training veterans for jobs in the maritime field, national defense education, militia training, and other national defense talent development. As an important recruiter that accounts for half of Quanzhou Municipality’s total directly-recruited NCOs, the institution has signed a strategic framework agreement with the PLA’s Eastern Theater Command Army. For more on the roles this institution plays in supporting national defense talent development, see: 福建省职业教育教学成果奖成果名称: 贯穿“军”元素高素质技术技能人才培养创新与实践 [“Fujian Provincial Vocational Education Teaching Achievement Award – Project Title: Innovation and Practice in Cultivating High-Quality Technical and Skilled Talents with Integrated ‘Military’ Elements”], 泉州海洋职业学院 [*Quanzhou Ocean Institute*], May 2024, p. 7; “浪花计划”退役军人培训班开班 [“The ‘Project Wavebreak’ Training Program for Retired Military Personnel has Commenced”], 泉州晚报 [*Quanzhou Evening News*], 16 October 2020, <https://www.qzoi.edu.cn/info/1351/31751.htm>.

<sup>132</sup> 我校获批第一批福建省退役军人思想政治教育基地 [“Our School Has Been Approved as One of the First Batch of Ideological and Political Education Bases for Retired Military Personnel in Fujian Province”], 泉州海洋职业学院 [*Quanzhou Ocean Institute*], 24 December 2023, <https://www.qzoi.edu.cn/info/1331/18501.htm>; 石狮百位民兵 参加海上集训 [“A Hundred Militia from Shishi Participated in Maritime Training Exercises”], 泉州晚报 [*Quanzhou Evening Post*], 8 August 2016, source no longer available online.

<sup>133</sup> For one example out of many, see these MSA experts discussing exam cheating and other bad practices from 2010. 郭金禄 [Guo Jinlu] and 黎志业 [Li Zhiye], 船员考试作弊预防 [“Prevention of Cheating in Seafarer Examination”], 中国海事 [*China Maritime Safety*], no. 4 (April 2010), pp. 52-60.

service records, crew members sometimes attempt to misuse the vessel's accounts to falsely report their presence on board a vessel during port entry and departure.<sup>134</sup> The consequences for these violations typically involve fines, suspension of certificates of competency for up to two years, as well as other penalties on individual applications for license upgrades and raises in grades. MSA inter-port office coordination has encountered difficulties in the verification of records, constraining efficiency in processing cases.<sup>135</sup>

On occasion, MSA branches will publicize law enforcement crackdown efforts on fraudulent practices. For example, in January 2024, an investigative team from the Dalian MSA discovered a case where one individual forged and altered another's certificate of competency to work aboard a ship as chief engineer when they were only qualified as a motorman. Due to the limits of the MSA's resources and authority in criminal prosecution, the Dalian MSA provided its evidence to the Dalian Public Security Bureau to establish a case against the perpetrators. The arrest unveiled an intermediary involving six individuals selling forged certificates.<sup>136</sup> Other regions have similarly reported a number of fraud cases. Recent intensified efforts by the Lianyungang MSA in the Spring of 2024 yielded discovery of up to 50 mariners who received certificates of competency using falsified sea service records.<sup>137</sup>

The MSA implements a demerit system to punish mariners who violate the rules, codified in the 2015 law "Measures of the People's Republic of China for the Administration of Scoring Illegal Acts of Seafarers" (《中华人民共和国船员违法记分办法》).<sup>138</sup> Every violation has a standard of value in points that is added to a merchant mariner's file during each scoring period (one year). If an individual accumulates 15 points during a period, the MSA shall withhold their certificates of competency, order them to participate in a five-day training on law and administrative regulation, and pass an examination. If they do not pass the examination, they may no longer serve aboard vessels. In more severe situations, wherein an individual surpasses the threshold twice within a scoring period or accumulates 40 points in two consecutive scoring periods, the MSA will also require them to take an

---

<sup>134</sup> 李国寅 [Li Guoyin] and 徐丹丹 [Xu Dandan], 国内船员服务资历的认定和管理 ["Recognition and Management of Service Qualifications for Seafarers Engaged in Domestic Voyages"], 中国海事 [China Maritime Safety], No. 7, 2024, pp. 14-16; 徐红儿 [Xu Hong'er], 赵海丽 [Zhao Haili], and 姜登运 [Jiang Dengyun], 海船船员任职资历审查问题及建议 ["Issues and Suggestions on Qualification Review of Seafarers"], 中国海事 [China Maritime Safety], No. 2, 2024, p. 31.

<sup>135</sup> Li and Xu, "Recognition and Management of Service Qualifications for Seafarers Engaged in Domestic Voyages," pp. 14-16.

<sup>136</sup> 海事公安携手，成功破获船员证书造假案 ["Maritime Safety Administration and Public Security Bureau Work Together to Successfully Crack Down on a Crew Certificate Forgery Case"], 中国海事 [China Maritime Safety], 17 December 2024, <https://www.xindemarinenews.com/china/57659.html>.

<sup>137</sup> 连云港海事局查处 6 名船员服务资历造假违法行为 ["Lianyungang Maritime Safety Administration Investigates and Punishes 6 Crew Members for Falsifying Their Service Qualifications"], 连云港海事局 [Lianyungang Maritime Safety Administration], 23 April 2024, <http://js.news.cn/20240423/ca7ac7d709eb4acc8b14f1485baba830/c.html>.

<sup>138</sup> The initial trial measures were first introduced in 2002 but were replaced by the 2015 version. This demerit system scores different violations according to standards established based on relevant laws, such as the Maritime Traffic Safety Law (《海上交通安全法》), the Regulation of the People's Republic of China on Seamen, the Regulations on the Safety Administration of Dangerous Chemicals (《危险化学品安全管理条例》), the Marine Environmental Protection Law (《海洋环境保护法》), and others. 中华人民共和国海事局关于印发《中华人民共和国船员违法记分办法》的通知 ["Notice of the Maritime Safety Administration of the People's Republic of China on the Issuance of the 'Measures of the People's Republic of China for the Administration of Scoring Illegal Acts of Seafarers'"], 中华人民共和国海事局 [Maritime Safety Administration of the People's Republic of China], 20 October 2015, <https://sea.dlou.edu.cn/2017/1102/c1583a68354/page.htm>.

assessment of their competency.<sup>139</sup> Violations do not necessarily signify the end of a seafarer's career but impose disciplinary actions to promote adherence to safety requirements. According to one assessment, the most common situations that cause seafarers to incur heavy point accrual fall under the following codes: 11001 (alcohol consumption), 11004 (leaving an accident), 11016 (overloading), 11017 (minimum safe manning), 21012 (operating outside of a vessel's navigational area – inland only), 11014 (not carrying certificates), and 11021 (failure to follow watchkeeping rules). These 8 or 15 point violations are the most common reasons for the loss of certificates.<sup>140</sup>

### *Crew Service Agencies*

Numerous crew service agencies (船员服务机构) operating across China provide services to Chinese seafarers. According to regulations on the administration of seafarer services, these are commercial entities “that act as an agent for mariners to handle relevant procedures such as applying for training, examinations, and obtaining certificates (including foreign crew certificates)” as well as businesses that “act as an agent for crew employers to manage crew affairs and provide manning for ships.” These agencies are administered and licensed by the MSA.<sup>141</sup> There are also numerous seafarer dispatch agencies (海员外派机构) licensed to provide manning and crew services to seafarers working on foreign-flagged vessels, subject to separate regulations.<sup>142</sup> Crew service agencies provide important services for seafarers navigating the complex experience, training, and qualification requirements for their professions, as well as providing ship management services for shipping enterprises, maintaining crew member files, and monitoring their status.<sup>143</sup>

Regulations state crew service agencies must meet certain standards to provide these services and not engage in illegal activities such as bribery, forgery, acting on behalf of non-qualified crew service agencies, and many other illegal activities.<sup>144</sup> The MSA may impose a fine of no less than 30,000 and no more than 150,000 RMB for agencies committing fraud, and in serious cases can suspend the agency's services for up to two years.<sup>145</sup> Despite the threat of fines and suspension, the industry is rife with illegal and fraudulent actors that cheat seafarers out of millions in earnings. They are further enabled by high levels of youth unemployment and desperate economic conditions, especially for people from poorer regions. Market conditions with periods of oversupply in seafarer labor worsen

---

<sup>139</sup> Ibid.

<sup>140</sup> “The Eight Common Situations Where Crew Members Receive High Demerit Points.”

<sup>141</sup> These agencies are categorized according to inland waterway vessel crew service agencies, Class A (甲级) - seagoing vessel crew service agencies (both international and domestic), and Class B (乙级) – seagoing vessel crew service agencies (domestic voyages only). See: Articles 2, 3, and 5, 中华人民共和国船员服务管理规定 [“Regulations of the People's Republic of China on the Administration of Seafarer Services”], 中国政府法制信息网 [China Government Legal Information Network], 9 February 2009, [https://www.moj.gov.cn/pub/sfbgw/flfggz/flfggzbmzg/200902/t20090209\\_144641.html](https://www.moj.gov.cn/pub/sfbgw/flfggz/flfggzbmzg/200902/t20090209_144641.html).

<sup>142</sup> 中华人民共和国海员外派管理规定 [“Regulations on the People's Republic of China on the Administration of Dispatched Seafarers”], 交通运输部发布 [Ministry of Transport Release], 20 September 2023.

<sup>143</sup> 中华人民共和国海事局关于进一步提升船员服务业务质量有关措施的通知 [“Notice of the Maritime Safety Administration of the People's Republic of China on Measures to Further Improve the Quality of Crew Services”], 中华人民共和国海事局 [Maritime Safety Administration of the People's Republic of China], 13 March 2023, <https://www.xindemarinenews.com/topic/chuanrenxinsheng/2023/0317/46456.html>.

<sup>144</sup> Ibid., Article 21.

<sup>145</sup> Article 59, “Regulation of the People's Republic of China on Seamen.” There are similar provisions in these regulations regarding illegal practices by shipping companies and seafarers, typically imposing a range of fines and license suspension.

the situation as seafarers, including captains, must rely on crewing agencies to find them positions aboard vessels, for steep fees. Agencies often make false promises regarding availability of work, vessel conditions, and pay. Many seafarers end up having to wait long periods to receive their pay from shipping companies and may have much of their pay withheld. In some cases, agencies fraudulently promise seafarers they can guarantee certificates within days, employment through inside connections with shipping companies, or guaranteed passing of examinations.<sup>146</sup> Some victims are promised quick entry into positions on merchant vessels, only to find themselves tricked into hard, ill compensated labor aboard fishing vessels.<sup>147</sup> It is unclear what the total impact these pernicious crew service agencies have on China's seafarer labor market, but this problem appears widespread and has caused victims of fraud to abandon merchant marine careers.<sup>148</sup>

Fighting fraud and abuse in the merchant marine and shipping companies is a perennial MSA task. The MSA promotes proper channels and procedures and attempts to improve seafarers' experiences by directing personnel toward verified service providers and other services listed on the MSA's online service platform.<sup>149</sup> It also publicly warns seafarers of the dangers lurking in crew service agencies, attempts to enforce the rules for advancement in licensing and training, and tries to protect the rights of seafarers.<sup>150</sup> In March 2023, the MSA introduced new measures that clarify the scope of permissible agency services to prevent fraudulent practices and false promises. The measures strengthen oversight of agency business filings and records, enhance transparency of crew service information, tighten regulations, and enforce stricter service quality standards.<sup>151</sup> Local MSA branches also combat fraud and misinformation online as many bad actors operate through virtual employment forums.<sup>152</sup>

---

<sup>146</sup> 薪水讨不回, 被黑中介骗 船长的下一步该怎么走? ["Unable to Recover Salary and Cheated by Black Agencies, What Should the Captain Do Next?"], 浙江新闻 [Zhejiang News], 21 April 2017, <https://news.csi.com.cn/0d27d1e6-c0ca-4796-901f-911c52d2a1f1.html>; 海事提醒! 船员遇到“黑中介,” 线索大胆报上来 ["MSA Reminder! Crew Members Should Report Any Evidence They Encounter as 'Black Agencies'"], 信德海事网 [Xinde Marine News], 11 July 2025, <https://www.xindemarinenews.com/china/60472.html>.

<sup>147</sup> For one example, see this case from 2016 of over 60 applicants being bussed from Shanghai to Rongcheng City, Shandong Province and made to work aboard small fishing vessels. They were promised positions aboard a large cargo ship with great pay. 应聘“高薪船员”竟被骗到山东荣成小渔船上做苦力 ["Applied for a 'High-Paying Crew Position' But was Tricked into Working as a Laborer on a Small Fishing Boat in Rongcheng, Shandong"], 龙 de 船人 [www.imarine.cn], 24 March 2016, <https://www.imarine.cn/6527.html>.

<sup>148</sup> There are many articles online with guides on where to avoid pitfalls, identify red flags, and how to find credible crew service agencies. 近期诈骗高发, 船员朋友请小心! ["There Has Been a Surge in Fraud Cases Recently, So Please Be Careful Fellow Crew Members!"], 中国海事 [China Maritime Safety], 26 November 2019, <https://www.xindemarinenews.com/topic/chuanrenxinsheng/2019/1126/16474.html>.

<sup>149</sup> MSA-verified crew service agencies are listed on the MSA's "One-Stop Maritime Service" (海事一网通办) under "Services / Dispatch Agency Search" (服务/外派机构查询), <https://zwfw.msa.gov.cn/dzzw/#/myQuery>, accessed 28 September 2025.

<sup>150</sup> Here is one guidance published by the Zhangzhou MSA in March 2023, outlining various fraudulent practices and ways seafarers can protect their rights and interests. 又是一年315, 船员朋友们这些诈骗“套路”你们都知道吗? ["It's March 15th Again This Year. My Fellow Crew Members, Do You Know These Fraudulent 'Tricks'?"], 漳州海事 [Zhangzhou MSA], 16 March 2023, <https://www.xindemarinenews.com/topic/chuanrenxinsheng/2023/0316/46429.html>.

<sup>151</sup> "Notice of the Maritime Safety Administration of the People's Republic of China on Measures to Further Improve the Quality of Crew Services."

<sup>152</sup> Here is one May 2022 example of the Guangdong MSA countering fraudulent advertising by a crew service agency regarding merchant marine physical examinations. 船员港湾 - 找“黑中介”办证要不得 ["Crew Harbor - Don't Use

## *Shipping Enterprises*

Shipping enterprises play a major role in maintaining safety and supporting the development of their workforces. They bear the responsibility of ensuring minimal manning of vessels is met and that crews are properly qualified.<sup>153</sup> China's top state-owned shipping conglomerate COSCO Shipping and its subsidiary COSCO Shipping Seafarer Management Co. Ltd. (中远海运船员管理有限公司) is the largest crew manning provider in China, providing management services for 42,000 seafarers working aboard the company's over 1,300 vessels.<sup>154</sup>

COSCO Shipping Seafarer Management is headquartered in Shanghai and has branches in Beijing, Guangzhou, Dalian, Tianjin, Qingdao, and Shenzhen. These branches provide essential workforce services to keep COSCO's ships manned by quality seafarers.<sup>155</sup> Just as COSCO Shipping forms the backbone of PRC merchant shipping across many sectors, COSCO Shipping Seafarer Management provides the workforce to keep those ships operating.<sup>156</sup> Not every shipping company has the resources and scale of COSCO Shipping, with its robust in-house crew service agency.

Outside of COSCO Shipping, there is a broader market-driven trend of shipping companies outsourcing personnel management services to crew service agencies to alleviate some administrative burdens. This has created problems in seafarer labor stability with high crew turnover rates. A major side effect has been problematic implementation of the ship Party branch system and the assignment of ship political commissars in companies that attempt to implement the commissar system.<sup>157</sup> In recent years, the government has tried to promote greater responsibility by shipping companies as well as in other industries concerning the safety and development of their workforces. Multiple official documents by the Ministry of Transport and MSA emphasized the core role of shipping companies in promoting workplace safety. These efforts sought to reduce the risk of accidents and mishaps at sea stemming from human factors, stressing that companies must have stronger safety management programs and strictly enforce standards in manning, qualifications, training, and watch standing.<sup>158</sup> In 2024, the MSA released a notice titled "Notice on Further Promoting Shipping

---

'Black Agencies' to Obtain Certificates"], 广东海事局官方澎湃号 [Official The Paper Account of the Guangdong MSA], 15 May 2022, [https://m.thepaper.cn/newsDetail\\_forward\\_18107024](https://m.thepaper.cn/newsDetail_forward_18107024).

<sup>153</sup> According to section 6 of the "Rules for the Examination and Licensing of Seafarers of the People's Republic of China," shipping companies must ensure their seafarers hold valid licenses and certificates, are familiar with their job duties, are familiar with the vessel's layout, characteristics, procedures, and limitations, and have the language and communication skills to perform their duties in emergencies and other situations. Article 44, "Regulation of the People's Republic of China on Seamen."

<sup>154</sup> The company's website claims it to be the world's largest crew management company. 公司简介 ["Company Introduction"], 中远海运船员管理有限公司 [COSCO Shipping Seafarer Management Co., Ltd.], Undated, [https://seafarer.coscoshipping.com/col10844/art/2017/art\\_10844\\_70268.html](https://seafarer.coscoshipping.com/col10844/art/2017/art_10844_70268.html), accessed 30 September 2025; 中远海运船员管理有限公司 ["COSCO Shipping Seafarer Management Co., Ltd."], 大连海事大学 - 就业信息网 [Dalian Maritime University - Employment Information Network], Undated, <https://myjob.dlmu.edu.cn/company/view/id/955371>, accessed 30 September 2025.

<sup>155</sup> Ibid.

<sup>156</sup> Company homepage, 中远海运船员管理有限公司 [COSCO Shipping Seafarer Management Co., Ltd.], undated, <https://seafarer.coscoshipping.com/>, accessed 30 September 2025.

<sup>157</sup> For one example of a state-owned shipping company that struggled with political commissar coverage over its fleet due to very low levels of in-house contracted seafarers, see the case of Guoneng Yuanhai Shipping Co. Ltd. (国能远海航运有限公司) in Kennedy, "Onboard Political Control," p. 8.

<sup>158</sup> The 2017 MSA notice instructed nationwide MSA branches to oversee shipping companies in vigorously promoting the implementation of the workplace safety individual responsibility system (航运公司全员安全生产责任制). Under this system, personnel at all levels are responsible and accountable for safe production. See 交通运输部海事局:全面加强航

Companies to Fulfill Their Principal Responsibilities for Seafarer Management” to urge companies to fulfill their roles in constructing high-quality seafarers and safety promotion.<sup>159</sup>

These efforts to increase the role of shipping companies in managing their workforce reflect a desire to stabilize the labor market and improve seafarer quality, mitigating some of the negative effects of relying on crew service agencies. This was one of the goals of the Ministry of Transport in carrying out the national goals of building a transportation power. It called for enterprises to contract directly with their own in-house crews (公司船员自有化).<sup>160</sup> According to the MSA’s 2024 notice, companies should assume responsibility for developing their own contracted seafarer workforce with the end goal being more stable crews, in-house rather than outsourced through manning agencies. As such, companies are encouraged to invest in career development and training programs for their workforces and foster lifelong seafarer careers.<sup>161</sup> According to accounts from 2021, around 80 percent of Chinese seafarers are “freelance seafarers” and exercise the freedom to choose and reject positions aboard ships, but also may not receive income or benefits when not at sea.<sup>162</sup> It is unclear how much the industry overall has shifted in line with policy goals, but major state-owned shipping companies are likely to be more compliant.

### Challenges Facing China’s Merchant Marine

The Chinese Seafarer Development Plan (2016-2020) openly admitted the troubles besetting the nation’s future merchant marine, including under-competitive pay and the lack of youth pursuing seafarer careers. The development plan promulgated several goals for promoting seafarer careers in support of national social and economic development goals.<sup>163</sup> The main goals covered issues across workforce structure in the seafarer labor market, training, education, and certification of seafarers, government and industry coordination, and improvements to the legal and regulatory system.<sup>164</sup> By 2023, some issues had been addressed, including improvements to the availability of information to seafarers, updates to industry regulations and standards, and successful compliance with IMO

---

运企业全员安全生产责任制 [“Maritime Safety Administration of the Ministry of Transport: Comprehensively Strengthening the All-Personnel Safe Production Responsibility System in Shipping Enterprises”], 信德海事网 [Xinde Marine News], 21 December 2017, <https://www.xindemarinenews.com/m/view.php?aid=1628>; 交通运输部办公厅关于进一步加强航运公司安全管理工作的意见 [“Opinions of the General Office of the Ministry of Transport on Further Strengthening the Safety Management of Shipping Companies”], 交通运输部网站 [Ministry of Transport Website], Document No. 2, 2019, [https://www.gov.cn/zhengce/zhengceku/2019-10/29/content\\_5446243.htm](https://www.gov.cn/zhengce/zhengceku/2019-10/29/content_5446243.htm).

<sup>159</sup> The Chinese name for this notice is 《关于进一步推动航运企业落实船员管理主体责任的通知》. See 如何落实船员管理主体责任? 航运企业看过来 [“How To Implement the Main Responsibility of Crew Management? Shipping Companies Should Take a Look”], 连云港海事 [Lianyungang Maritime Safety Administration], 24 February 2025, [https://www.sohu.com/a/863088468\\_121106832](https://www.sohu.com/a/863088468_121106832).

<sup>160</sup> “The Ministry of Transport, Ministry of Education, Ministry of Finance, Ministry of Human Resources and Social Security, Ministry of Veterans Affairs, and the All-China Federation of Trade Unions Issued Guiding Opinions on Strengthening the Development of High-Quality Seafarers.”

<sup>161</sup> Bringing crewing back in-house will also require these companies to have greater involvement in manning, certification, compliance with regulations, tracking and management of crew performance, and conducting assessments. Companies are also expected to be more involved in handling seafarer labor contracts, onboarding new hires, and managing seafarer benefits. “How To Implement the Main Responsibility of Crew Management? Shipping Companies Should Take a Look.”

<sup>162</sup> 为什么船员没有五险一金且休假后没有工资呢? 远洋船员告诉你答案 [“Why Do Seafarers Not Have the Five Insurances and Fund and No Salary After Taking Leave? Ocean-Going Seafarers Tell You the Answer”], 网易 - 航海大世界 [Netease - The World of Sailing], 12 October 2021, <https://www.163.com/dy/article/GM48KUE40552G72H.html>

<sup>163</sup> “Chinese Seafarer Development Plan (2016-2020).”

<sup>164</sup> See the six goals of the “Chinese Seafarer Development Plan (2016-2020).”

conventions. These are all areas relatively within the government's span of control. Addressing the seafarer profession's decline in popularity, fraud and abuse within the industry, and other social factors are likely much harder in practice. Promotion of the merchant marine and overall national maritime awareness may not be enough to shift the younger population's economic and social considerations when it comes to career choice.

The PRC will continue its effort to improve the industry where possible. In December 2023, the Ministry of Transport released "Guiding Opinions on Accelerating the High-Quality Development of the Modern Shipping Services Industry" to improve shipping industry finance, insurance, and legal services. It also called for strengthening support for seafarer talent, especially in new market sectors such as liquified natural gas.<sup>165</sup> An official explanation of the opinions stated, "in response to the problem that construction of China's seafarers needs to be strengthened (有待加强), the 'Guiding Opinions' put forward specific measures such as optimizing seafarer education and training and institutional management, and importing and increasing the supply of urgently needed talent."<sup>166</sup> Additionally, as mentioned previously, the PRC is encouraging shipping enterprises to play a greater role in cultivating seafarer talent and bringing them in-house to enhance labor stability and the attractiveness of the profession.<sup>167</sup>

## Conclusion

Beijing has a strong merchant marine commensurate with the scale and scope of its maritime industries.<sup>168</sup> The importance of this workforce to advancing its national economic and geostrategic goals cannot be overstated. China is currently in a period of accelerated development of national transportation networks as part of its long-term goal to become a "Transport Power" (交通强国), of which maritime shipping is a critical component.<sup>169</sup> At the same time, military-civil fusion efforts

---

<sup>165</sup> 交通运输部 中国人民银行 国家金融监督管理总局 中国证券监督管理委员会 国家外汇管理局关于加快推进现代航运服务业高质量发展的指导意见 ["The Ministry of Transport, People's Bank of China, State Financial Supervision and Administration Commission, China Securities Regulatory Commission, and the State Administration of Foreign Exchange Issued Guiding Opinions on Accelerating the High-Quality Development of the Modern Shipping Services Industry"], 交通运输部 [Ministry of Transport], Document No. 173, 2023, [https://xxgk.mot.gov.cn/jigou/syj/202312/t20231213\\_3967572.html](https://xxgk.mot.gov.cn/jigou/syj/202312/t20231213_3967572.html).

<sup>166</sup> 《关于加快推进现代航运服务业高质量发展的指导意见》解读 ["Interpretation of the 'Guiding Opinions on Accelerating the High-Quality Development of Modern Shipping Services'"], 交通运输部网站 [Ministry of Transport Website], 14 December 2023, [https://www.gov.cn/zhengce/202312/content\\_6920263.htm](https://www.gov.cn/zhengce/202312/content_6920263.htm).

<sup>167</sup> Ibid.

<sup>168</sup> Chinese news hails the strength of its maritime industry, stating it has the world's largest shipping fleet, has created a "world-class port cluster," and international shipping volume accounting for nearly one-third of the world's total. See: 王存福 [Wang Cunfu] and 叶昊鸣 [Ye Haoming], 我国国际海运量占全球近三分之一 ["China's International Shipping Volume Accounts for Nearly One-Third of the World's Total"], 新华社 [Xinhua], 11 July 2025, [https://www.gov.cn/yaowen/liebiao/202507/content\\_7031641.htm](https://www.gov.cn/yaowen/liebiao/202507/content_7031641.htm).

<sup>169</sup> The Ministry of Transport, National Railway Administration, Civil Aviation Administration of China, State Post Bureau, and China State Railway Group Co., Ltd. jointly issued a five-year plan of action entitled the "Five-Year Action Plan for Accelerating the Construction of a Transportation Power (2023-2027)" which clarified ideas, goals, and tasks for the current five-year plan. See: 刘昕 [Liu Xin], 贯彻落实党的二十大精神 谋划加快建设交通强国未来五年重点工作 - 《加快建设交通强国五年行动计划 (2023—2027 年)》解读 ["Implementing the Spirit of the 20th CCP National Congress and Planning Key Tasks for Accelerating the Development of a Transportation Power Over the Next Five Years - Interpretation of the 'Five-Year Action Plan for Accelerating the Development of a Transportation Power (2023-2027)']", 交通运输部网站 [Ministry of Transport Website], 23 April 2023, [https://www.gov.cn/zhengce/2023-04/23/content\\_5752770.htm](https://www.gov.cn/zhengce/2023-04/23/content_5752770.htm); 中共中央 国务院印发《交通强国建设纲要》 ["The CCP Central Committee and the State

nationwide are gradually improving the availability of various civilian commercial assets in support of PLA logistics. As regional tensions grow and the PRC continues to develop national defense mobilization capabilities, shipping enterprises and the merchant marine will likely see their involvement deepen.

As a professional workforce, China's merchant marine has overall kept compliance with major international standards, such as the IMO STCW. As such, the PRC's system for national seafarer management through the MSA has maintained a foundation of rules and regulations for training, education, licensing, and qualification that keeps PRC ships manned with competent personnel. To demonstrate the extent of PRC programs built to support this workforce, much of this report has explored and introduced the system within which PRC seafarers operate. Much of its content will be familiar to other international seafarers who may sometimes work alongside or in coordination with PRC crews on global waterways and ports. Some aspects are unique to China, including the scale of its state-owned shipping conglomerates and the CCP elements embedded within, including the presence of ship Party branches and the ship political commissar system.

However, China may be experiencing headwinds in long-term development of its merchant marine. The 2017 Chinese Seafarer Development Report stated in its conclusion that “the attractiveness of the seafarer profession has declined, the wage gap between ship and shore has narrowed, the social status of seafarers has declined, the training cycle is long, and the turnover rate is high. [These conditions] are making it difficult for seafarer quantity and quality to meet the demand of shipping development.”<sup>170</sup> While not unique to China, many of these problems impact China's economy acutely, as much of its inputs and outputs are dependent on global shipping.<sup>171</sup>

To remedy some of these problems, Beijing has embarked on a long-term effort to keep its shipping industry invigorated with a professional merchant marine that seeks to compete globally. This effort was embodied in the Chinese Seafarer Development Plan (2016-2020), which outlined many specific development goals. While some improvements have occurred, stubborn problems remain. Current efforts seek to improve the professional environment for PRC seafarers by strengthening MSA law enforcement, cleaning up the crew services market, and doing more to protect seafarer rights and interests.<sup>172</sup>

The PRC has achieved some success in cultivating a seafarer workforce that satisfies national strategic and shipping development needs. Standards and professionalism have likely risen at a minimum in-step with IMO requirements. The update and release of the “20 Rules” and the MSA's public information efforts have likely had a positive impact. However, shortages of qualified candidates for specific ship positions persist and the profession's lack of popularity with younger graduates could progressively undermine seafarer quality, as the best and brightest choose to pursue other careers. If shortages in junior officer positions continue, Beijing's development goals focused on strengthening seafarer officers may not be achieved. In terms of fostering in-house seafarer talent,

---

Council Issued the ‘Outline for Building a Transportation Power’], 新华社 [Xinhua], 19 September 2019, [https://www.gov.cn/zhengce/2019-09/19/content\\_5431432.htm](https://www.gov.cn/zhengce/2019-09/19/content_5431432.htm).

<sup>170</sup> “2017 Chinese Seafarer Development Report,” p. 57; These issues were detailed even further in the “Chinese Seafarer Development Plan (2016-2020).”

<sup>171</sup> According to World Bank data from 2024, China is the world's largest exporter of goods and services. See: “Exports of goods and services (current US\$),” World Bank Group, [https://data.worldbank.org/indicator/NE.EXP.GNFS.CD?most\\_recent\\_value\\_desc=true](https://data.worldbank.org/indicator/NE.EXP.GNFS.CD?most_recent_value_desc=true), accessed 16 October 2025.

<sup>172</sup> “2023 Chinese Seafarer Development Report,” pp. 19-22.

state-owned shipping conglomerates are likely the best performers in contracting directly with seafarers while many other enterprises are probably still outsourcing manning services. The PRC has a robust network of educational and training programs to enable talent development, but likely needs to strengthen MSA efforts in preventing fraud and abuse of the examination and licensing system. There are clear efforts across government, institutions, and industry to improve coordination to enable standardized and orderly functioning of the seafarer labor market, but these are more recent and will need time before results can be assessed. The transfer of demobilized PLA personnel into the merchant marine is a consistent line of effort to inject talent into the workforce. Ultimately, whether these trends improve or worsen will be observed in the active merchant marine figures contained in the annual seafarer development report and other industry reports. Overall, China's government and state-owned enterprises have made improvements within the span of their control but will need to vigorously address the downdrafts and headwinds threatening to blow the merchant marine off the ambitious course that Beijing has set for it.

## Appendix

Table of Seafarer Training Certificates, Codes, and Validity Periods.<sup>173</sup>

| Seafarer Training Certificate Programs                                                 | Code  | Max Validity Period |
|----------------------------------------------------------------------------------------|-------|---------------------|
| Basic Safety Training Certificate (基本安全培训合格证)                                          | Z01   | 5 years             |
| Proficiency in Survival Craft and Rescue Boats Certificate (精通救生艇筏和救助艇培训合格证)           | Z02   | 5 years             |
| Proficiency in Fast Rescue Boat Certificate (精通快速救助艇培训合格证)                             | Z03   | 5 years             |
| Advanced Fire Fighting Certificate (高级消防培训合格证)                                         | Z04   | 5 years             |
| Proficiency in Medical First Aid Certificate (精通急救培训合格证)                               | Z05   | Long-term           |
| Ship Medical Care Certificate (船上医护培训合格证)                                              | Z06   | Long-term           |
| Security Awareness Training Certificate (保安意识培训合格证)                                    | Z07   | Long-term           |
| Designated Security Duties Training Certificate (负有指定保安职责船员培训合格证)                      | Z08   | Long-term           |
| Ship Security Officer Training Certificate (船舶保安员培训合格证)                                | Z09   | Long-term           |
| Basic Oil & Chemical Tanker Cargo Operations Training Certificate (油船和化学品船货物操作基本培训合格证) | T01   | 5 years             |
| Advanced Oil Tanker Cargo Operations Training Certificate (油船货物操作高级培训合格证)              | T02   | 5 years             |
| Advanced Chemical Tanker Cargo Operations Training Certificate (化学品船货物操作高级培训合格证)       | T03   | 5 years             |
| Basic Liquified Gas Tanker Cargo Operations Training Certificate (液化气船货物操作基本培训合格证)     | T04   | 5 years             |
| Advanced Liquified Gas Tanker Cargo Operations Training Certificate (液化气船货物操作高级培训合格证)  | T05   | 5 years             |
| Passenger Ship Crew Special Training Certificate I (客船船员特殊培训合格证 I)                     | T06-1 | 5 years             |
| Passenger Ship Crew Special Training Certificate II (客船船员特殊培训合格证 II)                   | T06-2 | 5 years             |
| Passenger Ship Crew Special Training Certificate III (客船船员特殊培训合格证 III)                 | T06-3 | 5 years             |
| Special Training Certificate for Operating Large Vessels (大型船舶操纵特殊培训合格证)               | T07   | Long-term           |

<sup>173</sup> These training certificates are no longer effective once an individual passes their 65th birthday. 船员职业常识百问百答（2022 版）[“100 Questions and Answers on Seafarer Professional Knowledge (2022 Edition)”], 中国海事局 [China Maritime Safety Administration], June 2022, pp. 7-8.

| <b>Seafarer Training Certificate Programs</b>                                                                                  | <b>Code</b> | <b>Max Validity Period</b> |
|--------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------|
| Special Training Certificate for High-Speed Craft Crew (Fully Cushioned Hovercraft) (高速船船员特殊培训合格证 (全垫升气垫船))                    | T08-1       | 2 years                    |
| Special Training Certificate for High-Speed Craft Crew (Surface Effect Vessels) (高速船船员特殊培训合格证 (水面效应船))                         | T08-2       | 2 years                    |
| Special Training Certificate for High-Speed Craft Crew (Hydrofoil) (高速船船员特殊培训合格证 (水翼船))                                        | T08-3       | 2 years                    |
| Special Training Certificate for High-Speed Craft Crew (Monohull) (高速船船员特殊培训合格证 (单体船))                                         | T08-4       | 2 years                    |
| Special Training Certificate for High-Speed Craft Crew (Multi-Hull) (高速船船员特殊培训合格证 (多体船))                                       | T08-5       | 2 years                    |
| Special Training Certificate for Crew Loading Bulk Solid Hazardous and Harmful Materials on Ships (船舶装载散装固体危险和有害物质作业船员特殊培训合格证) | T09         | Long-term                  |
| Special Training Certificate for Crew Loading Packaged Hazardous and Hazardous Materials on Ships (船舶装载包装危险和有害物质作业船员特殊培训合格证)   | T10         | Long-term                  |
| Basic Training Certificate for Crew Aboard Ships Using Gas or Other Low-Flashpoint Fuels (使用气体或其他低闪点燃料船舶船员基本培训合格证)             | T11         | 5 years                    |
| Advanced Training Certificate for Crew Aboard Ships Using Gas or Other Low-Flashpoint Fuels (使用气体或其他低闪点燃料船舶船员高级培训合格证)          | T12         | 5 years                    |
| Basic Training Certificate for Vessels Operating in Polar Waters (极地水域船舶操作船员基本培训合格证)                                           | T13         | 5 years                    |
| Advanced Training Certificate for Vessels Operating in Polar Waters (极地水域船舶操作船员高级培训合格证)                                        | T14         | 5 years                    |

Application Requirements for Training, Qualifications, and Examinations for Seafarer Certificates of Competency<sup>174</sup>

| Position                               | Completion of Basic Safety & Professional Skills (Listed by Code) and Job-Specific Proficiency Training <sup>175</sup>                                                                               | Sea Service Experience (海上任职资历)                                                            |                                                                                                                                                                    | Examination                                    |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
|                                        |                                                                                                                                                                                                      | Sea Service Experience (海上服务资历) <sup>176</sup>                                             | On-Board Training (船上见习)                                                                                                                                           |                                                |
| <b>Captain / Chief Engineer</b>        | Z01, Z02, Z04, Z05, Z06 (only for captains on ships 500 gross tons or greater), Z07, Z08. Complete additional corresponding captain / chief engineer job-specific proficiency training.              | Serve as chief mate / second engineer for 18 months.                                       | Complete trainee period of not less than 3 months on the corresponding class of ship and navigational area.                                                        | Pass the captain / chief engineer exam.        |
| <b>Chief Officer / Second Engineer</b> | Z01, Z02, Z04, Z05, Z06 (only for chief officers on ships 500 gross tons or greater), Z07, Z08. Complete additional corresponding chief officer / second engineer job-specific proficiency training. | Serve as second officer / third engineer for 12 months.                                    | Complete trainee period of not less than 3 months on the corresponding class of ship and navigational area.                                                        | Pass the chief officer / second engineer exam. |
| <b>Second Officer / Third Engineer</b> | Z01, Z02, Z04, Z05, Z07, Z08. Exempt (免除) from additional job-specific proficiency training.                                                                                                         | Serve as third officer / fourth engineer for 12 months.                                    | No trainee period required.                                                                                                                                        | No exam required.                              |
| <b>Third Officer / Fourth Engineer</b> | Z01, Z02, Z04, Z05, Z07, Z08. Complete additional corresponding third officer / fourth engineer job-specific proficiency training.                                                                   | Serve as RFPNW / RFPEW / or AB-D / AB-E for a cumulative period of no less than 18 months. | Perform bridge or engine room watch duties for not less than 6 months on a vessel of corresponding class under the guidance of the captain or a qualified officer. | Pass the third officer / fourth engineer exam. |

<sup>174</sup> These requirements are attached to the “Regulations on Examination and Certification of Competence for Seafarers of the People's Republic of China.”

<sup>175</sup> Ship officers from captains / chief engineers to third officers / fourth engineers serving aboard vessels less than 500 gross tons or 750 kw power plants (except special types of vessels) are exempted from survival craft and rescue boat proficiency training, advanced firefighting training, and first aid proficiency training. Ibid., p. 6.

<sup>176</sup> The term “sea service experience” (海上服务资历) refers to experience in the corresponding navigational area, grade, and position obtained within five years before participating in job-specific proficiency training. There are a number of sea time related rules related to grade, tonnage/horsepower, and navigational areas. For example, applying for a raise of grade with an unlimited navigational area certificate requires at least six months of sea service on ships in the corresponding navigational area, with the remaining sea time allowed to be aboard ships in coastal navigational areas. Ibid., pp. 6-8.

|                                                                                             |                                                                                        |                                     |                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                      |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Able Seafarer - Deck / Able Seafarer - Engine</b>                                        | Z01, Z02, Z07, Z08. No additional job-specific proficiency training required.          | Serve as AB-D / AB-E for 18 months. |                                                                                                                                                                                                                                                                                                                                                                   | Pass the AB-D / AB-E exam.                                                                                           |
|                                                                                             |                                                                                        | Serve as AB-D / AB-E for 12 months. | Complete onboard traineeship for no less than 3 months within the last 6 months in accordance with the requirements of a training plan and record.                                                                                                                                                                                                                | No exam required.                                                                                                    |
| <b>RFPNW / RFPEW</b>                                                                        | Z01, Z07, Z08. Complete corresponding RFPNW / RFPEW job-specific proficiency training. |                                     | Complete no less than 6 months of seagoing service on the corresponding class of vessel, of which 3 months should be spent performing watchkeeping duties under the direct supervision of a qualified officer or qualified support-grade crew member, or complete 3 months of onboard training in accordance with the requirements of a training plan and record. | Pass the RFPNW / RFPEW exam.                                                                                         |
| <b>Ratings not qualified to participate in a navigational or engine watch (Example: OS)</b> | Z01, Z07.                                                                              |                                     |                                                                                                                                                                                                                                                                                                                                                                   | Pass the Z01 and Z07 proficiency exams. Foreign language test required if applying to work on international voyages. |

## **About the Author**

Conor Kennedy is an affiliate with the China Maritime Studies Institute. He previously served as an assistant professor at the Naval War College's China Maritime Studies Institute. His research focuses on the People's Republic of China's maritime and military development. He holds a master of arts degree in international studies from the Johns Hopkins University School of Advanced International Studies (Hopkins-Nanjing Center), and two bachelor of arts degrees in political science and Chinese language from the University of Massachusetts Amherst. He is the recipient of the National Security Education Program's David L. Boren Fellowship to China, 2013–14.

*The views and opinions contained in this report are the authors alone. They do not necessarily represent the perspectives of the Department of War or the Naval War College.*